

Anticipating demand in carbon-aware scheduling across data centres: guarantees and fair carbon attribution

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Abstract

Shifting deferrable jobs to cleaner hours and locations cuts data-centre emissions, but online schedulers act before future grid conditions and jobs are known. We formulate multi-site, deadline-constrained scheduling of bag-of-tasks jobs as a minimum-cost flow problem and study CARMA (Carbon-aware Anticipatory Receding-horizon Multi-site Allocation), which adds “ghost jobs” for expected arrivals to each hourly plan. Uncertainty about demand alone forces any online algorithm, even with perfect carbon forecasts, to a competitive ratio of $\Omega(\sqrt{\theta})$, where θ is the ratio of highest to lowest unit emission cost. Under explicit assumptions (a backstop resource, episodes within the planning window), CARMA never misses a deadline and is optimal under exact predictions; these assumptions do not hold in the weekly experiments. On 2024 regional grid data for Great Britain, CARMA comes within 8.0–9.8% of the clairvoyant optimum, against 15.5–16.3% for the best baseline without demand information, and cuts attributional emissions by 37–69% relative to carbon-agnostic operation. A scenario-based stochastic lookahead with the same demand information is 0.3–1.0 points better but needs 12–37 times more computing time. On a trace-derived workload with a biased demand profile, CARMA’s gap is 21.2% against 27.5–36.8%. Anticipating demand is worth more than perfect carbon foresight in these settings. Under a stylised marginal-emission proxy, however, achievable reductions shrink to 14% and CARMA’s edge becomes small. Dual prices of the pooled problem give a carbon attribution in the core of the hindsight pooling game, whereas origin- and host-based accounting violate coalitional stability in every week.

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1. Introduction

Data centres are among the fastest-growing electricity consumers. Efficiency gains kept their energy use nearly flat through the 2010s even as computing output grew several-fold [1, 2]. That buffer is now thinning, and information and communication technology accounts for an estimated 2.1–3.9% of global greenhouse-gas emissions [3]. Large machine-learning workloads have drawn particular attention [4, 5, 6]. The carbon intensity of grid electricity varies by a factor of ten or more across regions and hours, driven by the output of wind, solar and dispatchable plant [7]. Operators can therefore cut emissions without new hardware by moving flexible work to cleaner places and times. Such *carbon-aware computing* [8, 9] is an operations problem: capacity is finite, deadlines bind, and both the grid and the job stream are uncertain.

Batch jobs such as model training, analytics, rendering and backups are a natural target, because they tolerate some delay. Previous work has shown the potential of temporal shifting (running later) [10, 11, 12] and of spatial shifting (running elsewhere) [13, 14]. It has also shown their limits. Sukprasert et al. [15] found that practical reductions from spatio-temporal shifting fall well short of idealised bounds. Most methods are evaluated with a scheduler that reacts to the jobs it has already seen, using point forecasts of carbon intensity. Three questions have received less attention. First, how much is lost because an online scheduler does not know which jobs will arrive next, and can any scheduler guarantee good performance in the worst case? Second, does it pay to price the uncertainty of carbon-intensity forecasts into scheduling decisions? Third, when several sites pool their capacity, how should the resulting emissions be attributed so that every participant is better off than on its own?

This paper addresses these questions with a scheduling policy, guarantees under explicitly stated assumptions, and a controlled evaluation on three years of regional grid data for Great Britain. Its building blocks (receding-horizon planning, network flows, conformal calibration and dual cost allocation) are standard; the contribution lies in their combination for this problem, the lower

bound, the conditions under which the guarantees hold and the empirical evidence. It makes five contributions.

1. **Formulation.** We state multi-site, deadline-constrained, width-limited scheduling of deferrable bag-of-tasks jobs, including the emissions of data transfer, as a minimum-cost flow problem (Proposition 1). Offline clairvoyant optima and every online planning step can therefore be solved to optimality, after quantisation, by fast combinatorial min-cost flow algorithms. We also prove that myopic receding-horizon scheduling can be arbitrarily worse than the clairvoyant optimum even with perfect carbon forecasts (Proposition 2).
2. **Algorithm.** We propose CARMA (Carbon-aware Anticipatory Receding-horizon Multi-site Allocation). At every hour, CARMA adds *ghost jobs* to its planning network. These represent expected arrivals, learned from the operator’s history, so the planner reserves scarce low-carbon capacity for work that has not yet arrived. The planning problem stays a single min-cost flow and decisions take tens of milliseconds.
3. **Guarantees.** We prove that uncertainty about demand alone forces every online algorithm, even one with perfect carbon forecasts, to a competitive ratio of $\Omega(\sqrt{\theta})$, about $\sqrt{\theta}/2$ for large θ , where θ is the ratio of the highest to the lowest unit emission (Theorem 1). In an idealised setting with a backstop resource, CARMA never misses a deadline whatever its predictions, so it inherits the θ -competitiveness that every on-time policy has. If, in addition, each episode fits within the planning window, CARMA is optimal when its predictions of demand and carbon are exact, and its excess emissions grow at most linearly in the prediction errors (Proposition 3 and Theorem 2). These assumptions do not hold in our weekly experiments, so we verify the theorem separately in episodes that satisfy them. We also show that the dual prices of the pooled scheduling network give a carbon attribution among sites that lies in the core of the pooling game, so no group of sites gains by leaving (Theorem 3).
4. **Forecasting and recalibration pipeline.** As a tool for the scheduling study, we build a reproducible 1–24-hour carbon-intensity forecaster from a public feed: direct gradient-boosted models with horizon-wise normalised conformal quantiles and adaptive recalibration, which keeps empirical coverage close to target under the decarbonisation drift of

2024. Its quantiles let us test risk-adjusted carbon pricing inside the scheduler. We do not claim a new forecasting method.

5. **Evidence.** Across 52 out-of-sample weeks with three workload traces each, three load levels, two operating regimes and a trace-driven workload from the Alibaba 2018 cluster trace, we compare CARMA with seven implementable online baselines, including a revenue-management protection-level policy and a scenario-based stochastic lookahead that uses the same demand information, with clairvoyant and ablated variants, and with the clairvoyant oracle. The results separate the value of anticipating demand from the value of better carbon forecasts, show when each matters, characterise how performance degrades with prediction errors, and show that origin- and host-based carbon accounting violate coalitional stability in every test week, whereas the dual attribution satisfies it exactly for the hindsight benchmark and in over 99% of the checks for CARMA’s realised emissions, where the adjusted attribution is a heuristic outside the theorem.

The results carry an operational message. Once spatial shifting is allowed, the dominant loss comes from demand myopia, not from forecast error. Anticipating demand cuts the gap of myopic planning to the clairvoyant optimum by 51–60% in the synthetic experiments, and by 42% on a trace-derived workload whose demand profile is learned from different days. Perfect carbon foresight or risk-averse carbon pricing adds little by comparison. A scenario-based stochastic lookahead with the same demand information is slightly better than CARMA but 12–37 times slower, and it does not retain the network-flow structure that our guarantees use. Without migration, temporal shifting alone achieves much smaller reductions, and near saturation it trades emissions against lateness. We also show the limits of the evidence. The headline reductions are attributional. Under a stylised marginal-emission proxy within the single GB market, most of them would be reallocations, and physical savings depend on low-carbon supply that would otherwise be curtailed. The rest of the paper is organised as follows. Section 2 reviews related work. Section 3 presents the model, the forecaster, CARMA and the experimental design. Section 4 establishes the theoretical guarantees, with proofs in Appendix A. Section 5 reports results, Section 6 discusses implications and limitations, and Section 7 concludes.

105 2. Related work

106 *Carbon-aware workload shifting.* Early green data-centre schedulers matched
 107 deferrable jobs to on-site solar production, for example GreenSlot [16] and
 108 GreenHadoop [11]. Other work exploited electricity-price and carbon differ-
 109 ences across sites through geographical load balancing [17, 13] and dynamic
 110 right-sizing [18]. With public carbon-intensity feeds, attention moved to grid
 111 emissions. Wiesner et al. [10] quantified the reductions from temporal shift-
 112 ing in several regions, including Great Britain, and proposed forecast-based
 113 lowest-window scheduling. Google’s carbon-intelligent compute system sets
 114 next-day virtual capacity curves from carbon forecasts and forecasts of the
 115 fleet’s own demand [8]. It is the closest industrial precedent for using demand
 116 forecasts, but it shapes hourly virtual capacity curves per cluster rather
 117 than planning individual jobs across sites. Cucumber admits delay-tolerant
 118 work only when forecast renewable supply allows it [19], and Caribou shifts
 119 serverless functions between regions [20]. CarbonScaler varies the allocation
 120 of elastic jobs with carbon intensity [12]. Ecovisor exposes the energy system
 121 to applications [21]. Carbon Explorer weighs scheduling against investment
 122 in renewables and storage [22]. Load migration between data centres can
 123 absorb curtailed renewable generation [14], and data-centre demand response
 124 has been surveyed in [23]. Sukprasert et al. [15] bound the potential of spatio-
 125 temporal shifting across 123 regions and find that simple policies capture
 126 most of the achievable benefit. Our study complements theirs by identifying
 127 *which* information limits online schedulers when capacity at clean sites is
 128 scarce.

129 *Online algorithms and guarantees.* Worst-case analysis of carbon-aware
 130 scheduling builds on online search and one-way trading [24], for which
 131 threshold policies achieve optimal competitive ratios. Lechowicz et al.
 132 derived optimal double-threshold algorithms for the online pause-and-resume
 133 problem [25]. With CarbonClipper, they also derived learning-augmented
 134 algorithms with an optimal consistency–robustness trade-off for spatiotem-
 135 poral allocation of a single deadline-constrained workload over a network
 136 of data centres [26]. These works model uncertainty about future carbon
 137 *prices* for one job. We study many jobs that compete for finite low-carbon
 138 capacity, and we show that uncertainty about *demand* creates a distinct
 139 lower bound that carbon forecasts cannot remove. Our guarantees follow
 140 the consistency–robustness paradigm of algorithms with predictions [27].

141 CARMA’s re-solving of a deterministic plan with expected future demand
142 is an instance of the certainty-equivalent re-solving heuristics of network
143 revenue management [28]. Our contribution is not the device itself but
144 its formulation as a single flow problem for carbon-aware scheduling, the
145 guarantees and the empirical attribution of value. Table 1 positions the
146 closest work along the dimensions that matter here.

Table 1: Positioning against the closest work, as modelled in the cited studies ($\checkmark$: explicitly modelled or provided; $-$: not modelled).

Work	Many jobs share finite capacity	Information on future demand	Worst-case guarantee	Attribution among sites
Lowest-window temporal shifting [10]	–	–	–	–
Geographical load balanc- ing [13]	$\checkmark$	–	–	–
Carbon-intelligent compute system [8]	$\checkmark$	$\checkmark$	–	–
Online pause-and- resume [25]	–	–	$\checkmark$	–
CarbonClipper [26]	–	–	$\checkmark$	–
Flow models of preemptive scheduling [29, 30]	$\checkmark$ (offline)	–	–	–
This paper	$\checkmark$	$\checkmark$	$\checkmark$ (A1–A2)	$\checkmark$

147 Our fairness result draws on cooperative game theory: linear production
148 games and flow games have non-empty cores that can be computed from dual
149 prices [31, 32].

150 *Energy-aware scheduling in operations.* Energy-efficient scheduling is a mature
151 topic in manufacturing operations [33], and green scheduling models have ap-
152 peared in this journal, for example for parallel machines with job splitting [34]
153 and for cloud resource allocation [35]. Most such models are deterministic and
154 solved offline, often with metaheuristics. We instead exploit exact network-
155 flow structure and embed it in an online lookahead policy. Network-flow
156 formulations of preemptive scheduling with release dates and deadlines go
157 back to Horn [29] and Federgruen and Groenevelt [30]. Proposition 1 extends
158 this classical construction to multiple sites, job widths, time-varying emission
159 costs and transfer emissions [36, 37]. In the taxonomy of Powell [38], CARMA
160 is a deterministic lookahead policy with a modified (anticipatory) model, in
161 the spirit of model predictive control [39, 40].

162 *Carbon-intensity forecasting.* Forecasts of grid carbon intensity use statistical
163 decomposition [41], machine learning with many exogenous drivers [42], day-
164 ahead regression for building demand response [43], and hierarchical models of
165 the generation mix, as in CarbonCast [44]. For Great Britain, NESO publishes
166 regional forecasts through its Carbon Intensity API [45]. These are average
167 (attributional) intensities. Marginal emission factors, which describe the plant
168 that responds to a change in load, can differ substantially from averages and are
169 the relevant measure for the system-level effect of load shifting [46, 47]. These
170 studies focus on point accuracy. For decisions, calibrated uncertainty matters
171 as well. Conformal prediction [48, 49, 50] provides distribution-free intervals
172 and has been extended to quantile regression [51], to multi-horizon [52] and
173 dynamic time series [53], and to arbitrary distribution shift through adaptive
174 conformal inference [54]. We use these tools to build a recalibrated forecaster
175 and to test, in a controlled way, whether risk-adjusted carbon costs improve
176 scheduling. The question is motivated by the optimizer’s curse [55] and by the
177 robust-optimisation view that point-estimate solutions can be fragile [56, 57].

178 3. Material and methods

179 3.1. Problem setting

180 We consider an operator that runs a set $\mathcal{S}$ of data-centre sites connected
181 by a wide-area network. Time is divided into hourly slots $t = 0, 1, 2, \dots$
182 Deferrable batch jobs arrive online. Job j is released at the start of hour a_j
183 at origin site $o_j \in \mathcal{S}$. It carries w_j server-hours of work made of independent
184 tasks (a bag of tasks), can use at most m_j servers concurrently, and must
185 finish before the start of hour $d_j > a_j$. Work may be paused and resumed,
186 and any part of it may run at a site other than its origin. Remote execution
187 requires moving input data. At site s and hour t , $K_s(t)$ servers are free for
188 batch work once the interactive load has been served.

189 Let $I_s(t)$ be the average carbon intensity of the electricity supplied to site
190 s during hour t (gCO₂/kWh). Executing one server-hour of job j at site s
191 during hour t emits

$$c_{o_j s}(t) = e \text{PUE}_s I_s(t) + \mathbf{1}[s \neq o_j] \delta \eta \frac{1}{2} (I_{o_j}(t) + I_s(t)), \quad (1)$$

192 where e is the IT energy per server-hour, PUE_s is the power usage effectiveness
193 of site s , δ is the data volume moved per migrated server-hour and η is the
194 energy intensity of network transfer (kWh/GB). The transfer term charges

195 network energy at the mean intensity of the two endpoints. With decision
 196 variables $x_{js}(t) \geq 0$ (server-hours of job j run at site s in hour t), the offline
 197 version of the problem is the linear programme

$$\min_{x, u \geq 0} \sum_j \sum_s \sum_{t=a_j}^{d_j-1} c_{o_js}(t) x_{js}(t) + \sum_j \Pi u_j \quad (2)$$

$$\text{s.t.} \quad \sum_s \sum_{t=a_j}^{d_j-1} x_{js}(t) + u_j = w_j \quad \forall j, \quad (3)$$

$$\sum_s x_{js}(t) \leq m_j \quad \forall j, t \in [a_j, d_j), \quad (4)$$

$$\sum_{j: a_j \leq t < d_j} x_{js}(t) \leq K_s(t) \quad \forall s, t, \quad (5)$$

198 where u_j is work left unfinished at the deadline and Π is a lateness penalty
 199 much larger than any emission cost. When migration is not allowed, $x_{js}(t) = 0$
 200 for $s \neq o_j$; these variables are removed from the problem. With complete
 201 knowledge of arrivals and intensities, the optimum of (2)–(5) is a lower bound
 202 on the emissions of every schedule that serves all jobs by their deadlines, and
 203 we use it as the *clairvoyant oracle*. When emissions are evaluated only for a
 204 subset of jobs (Section 3.5), the oracle still has to serve every job under the
 205 same constraints but is charged emissions only for the evaluated ones, so that
 206 it remains a lower bound on the evaluated emissions of any schedule that
 207 meets all deadlines. Online policies that finish some work late lie outside this
 208 set; Section 3.5 explains how we compare them. An online scheduler has to
 209 fix $x_{js}(t)$ at the start of hour t . At that point it knows the jobs released so
 210 far, the current intensities $I_s(t)$ and nothing more than forecasts of $I_s(t+h)$
 211 and of future arrivals.

212 **Proposition 1** (Network-flow structure). *Problem (2)–(5) is a minimum-*
 213 *cost flow problem on a network with a source, one node per job, one node*
 214 *per (job, hour) pair, one node per (hour, site) pair and a sink. The source*
 215 *supplies $\sum_j w_j$ units of flow and the sink absorbs them. Arc capacities are w_j*
 216 *(source $\rightarrow j$), m_j ($j \rightarrow (j, t)$) and $K_s(t)$ ($(t, s) \rightarrow \text{sink}$). The only arcs with a*
 217 *cost are $(j, t) \rightarrow (t, s)$, priced at $c_{o_js}(t)$, and a slack arc $j \rightarrow \text{sink}$ priced at Π .*

218 *Proof.* Because the supply equals $\sum_j w_j$ and arc source $\rightarrow j$ has capacity w_j ,
 219 every such arc is saturated, and conservation at the job node is Eq. (3). The
 220 capacity of arc $j \rightarrow (j, t)$ is Eq. (4), conservation at (j, t) splits that hour's
 221 work across sites, and the capacity of $(t, s) \rightarrow \text{sink}$ is Eq. (5). Conversely,
 222 every feasible flow defines a feasible (x, u) with the same cost. The constraint

matrix of the flow formulation is a node-arc incidence matrix, in which every column has exactly one $+1$ and one -1 , so it is totally unimodular. $\square$

Proposition 1 has a practical consequence. The oracle and every planning problem solved online (Section 3.3) can be solved by combinatorial min-cost flow algorithms [36], which return integral optimal flows for integral data. We quantise work, widths and capacities to 0.01 server-hours (widths rounded up, capacities down) and costs to 0.01 gCO₂, and solve the quantised network to optimality with the min-cost flow solver of OR-Tools [58] (version 9.15, a cost-scaling push-relabel algorithm). The solutions are therefore optimal for the quantised problem, and the simulator treats remaining work below 0.05 server-hours as complete; it checks every executed allocation against the capacity and width limits up to this tolerance. On the weekly clairvoyant problems (1,100–2,800 jobs), the flow solver returned the same objective as the LP solver HiGHS on the unquantised problem to within a relative 5×10^{-5} and was 1.4–3.5 times faster (median 2.3; 0.3–1.2 s against 0.4–3.8 s; script `bench_solvers.py`; one CPU core). The flow time includes two solves, for the emissions oracle and for the service-adjusted oracle, whereas the LP time is for one solve.

3.2. Probabilistic carbon-intensity forecasting

Point model. For each horizon $h = 1, \dots, H$ ($H = 24$ h) we fit a direct gradient-boosted regression tree model [59] that predicts the change $I_s(t + h) - I_s(t)$ from information available at issue time t . Direct per-horizon models avoid the error accumulation of recursive strategies [60]. One model per horizon is pooled across sites, with the site as a categorical feature. The features are the current intensity; differences between the current value and lags of 1, 2, 3, 6, 12 and 23 hours; the value observed at the same hour of the target on the previous day and in the previous week; a 24-hour rolling mean and mean absolute change (volatility); the national (GB) intensity; the regional wind, solar and gas shares; and calendar encodings of the target time (hour of day, day of week, annual sine and cosine). Models are fitted with absolute-error loss, so they estimate the conditional median. We use scikit-learn’s histogram gradient boosting [61] with 400 iterations, learning rate 0.05, 31 leaves and at least 50 samples per leaf. The models use only information available when the forecast is issued (gaps in the data are filled causally; Section 3.5), and no weather forecasts. This is deliberate: the goal is a forecaster any operator can build from a public intensity feed.

259 *Conformal upper quantiles.* A point forecast $\hat{I}_s(t+h|t)$ is turned into cali-
 260 brated upper quantiles by split conformal prediction [48, 49] with normalised
 261 scores. On a calibration window disjoint from training, we compute for each
 262 site s and horizon h

$$r_{s,h}(t) = \frac{I_s(t+h) - \hat{I}_s(t+h|t)}{\sigma_s(t) + \epsilon}, \quad (6)$$

263 where $\sigma_s(t)$ is the 24-hour mean absolute hourly change observed at issue
 264 time and $\epsilon = 5$ gCO₂/kWh. Only pairs whose target $t+h$ also lies in the
 265 calibration window are used, so that no calibration outcome falls in the
 266 tuning period. For a target level β , the upper forecast is $U_s^\beta(t+h|t) =$
 267 $\hat{I}_s(t+h|t) + \hat{q}_{s,h}(\beta_{s,h}(t))(\sigma_s(t) + \epsilon)$, where $\hat{q}_{s,h}(\beta)$ is the $\lceil (n+1)\beta \rceil$ -th smallest
 268 of the n calibration scores, the split-conformal quantile. Grid carbon intensity
 269 is not exchangeable over time: 2024 was markedly cleaner than 2022–2023
 270 (Fig. 3). We therefore update the working level online by adaptive conformal
 271 inference (ACI) [54]. Once $I_s(t+h)$ is observed,

$$\alpha_{s,h} \leftarrow \alpha_{s,h} + \gamma((1-\beta) - \mathbf{1}[I_s(t+h) > U_s^\beta(t+h|t)]), \quad \beta_{s,h}(t) = 1 - \alpha_{s,h}, \quad (7)$$

272 with step size $\gamma = 0.005$. At every hour, the outcomes observed in that
 273 hour update the levels before the new forecasts are issued, so feedback for
 274 horizon h arrives h hours after the forecast. When the working level reaches
 275 1, the bound is set to the largest intensity of the site in the training period,
 276 a finite worst case in place of the infinite interval of ACI; when it reaches
 277 0 the bound is zero, and bounds are floored at zero. Gap-filled outcomes
 278 give no feedback. For immediate feedback and unbounded intervals, ACI
 279 guarantees that the long-run miscoverage frequency converges to $1 - \beta$ for
 280 any data sequence [54]. Our variant has delayed feedback, a finite ceiling and
 281 a quantile grid of step 0.001, so we do not claim this guarantee and use ACI
 282 as an adaptive recalibration whose coverage is checked empirically (Appendix
 283 B). The level β then acts as a risk dial that remains roughly calibrated under
 284 drift, and with $\beta = 0.5$ the procedure removes level shifts in the point model
 285 (a recalibrated median).

286 3.3. CARMA: carbon-aware anticipatory receding-horizon multi-site allocation

287 A standard receding-horizon (model predictive control, MPC) sched-
 288 uler [39] solves, at every hour t , the flow problem of Proposition 1 restricted
 289 to pending jobs and the window $[t, t+H)$. Future intensities are replaced by

forecasts, and only the first hour of the plan is executed. Such a planner is *myopic about demand*: it sees only jobs that have already arrived. The next result shows that this can be arbitrarily costly even when carbon intensities are known exactly.

Proposition 2 (Myopia can be unboundedly costly). *Even with perfect intensity forecasts, myopic receding-horizon scheduling has no finite competitive ratio with respect to the clairvoyant oracle.*

Proof. Take a clean site A with one server and a dirty site B with ample capacity. Let $c_A(0) = \varepsilon$, $c_A(1) = 0$ and $c_B(0) = c_B(1) = c$, and ignore transfer costs. A flexible job ($w = 1$, $m = 1$) arrives at $t = 0$ with deadline $d = 2$. An urgent job ($w = 1$) arrives at $t = 1$ with $d = 2$. At $t = 0$ the myopic planner sees only the flexible job and defers it to A at $t = 1$ (cost $0 < \varepsilon$). At $t = 1$ both jobs must run in the same hour, one of them at B , so the total cost is c . The oracle runs the flexible job at $t = 0$ and the urgent job at $t = 1$, both at A , for a total of ε . The ratio c/ε is unbounded as $\varepsilon \rightarrow 0$. $\square$

The mechanism in Proposition 2 is common in practice. Low-carbon capacity is scarce, and a myopic planner spends it on flexible work that could have waited or run elsewhere, leaving urgent jobs that arrive later to run on carbon-intensive capacity. CARMA addresses it by anticipating demand inside the same flow problem, and optionally prices forecast risk.

Anticipated demand (ghost jobs). From the operator’s own history we estimate, for every hour of the week τ , origin o and window class b , the expected work $\bar{W}_{\tau ob}$ and the expected total width $\bar{M}_{\tau ob}$ of arriving jobs. Window classes are the bins $\mathcal{B} = \{1, 2, 4, 6, 9, 13, 18, 24\}$ hours. A job is assigned to the largest bin not exceeding its window, so ghost jobs are never more flexible than the jobs they represent. Because binning shortens windows, the width of each ghost job is raised where necessary to at least its work divided by its window, which makes every ghost job window-feasible. At hour t , for each $k = 1, \dots, \ell$ (the look-ahead), CARMA adds to the planning problem a *ghost job* for every (o, b) with work $\kappa \bar{W}_{\tau(t+k)ob}$, width $\kappa \bar{M}_{\tau(t+k)ob}$, release $t + k$ and deadline $\min(t + k + b, t + H)$. Ghost jobs compete with real jobs for capacity, but their slack arc is priced at a moderate penalty Π_g , set to 1.5 times the largest local unit cost in the window, rather than at Π . They can therefore reserve clean capacity when it is economical but never force real work to be late. The weight $\kappa \geq 0$ scales how much of the expected demand is anticipated. With $\kappa = 0$, CARMA reduces to myopic MPC.

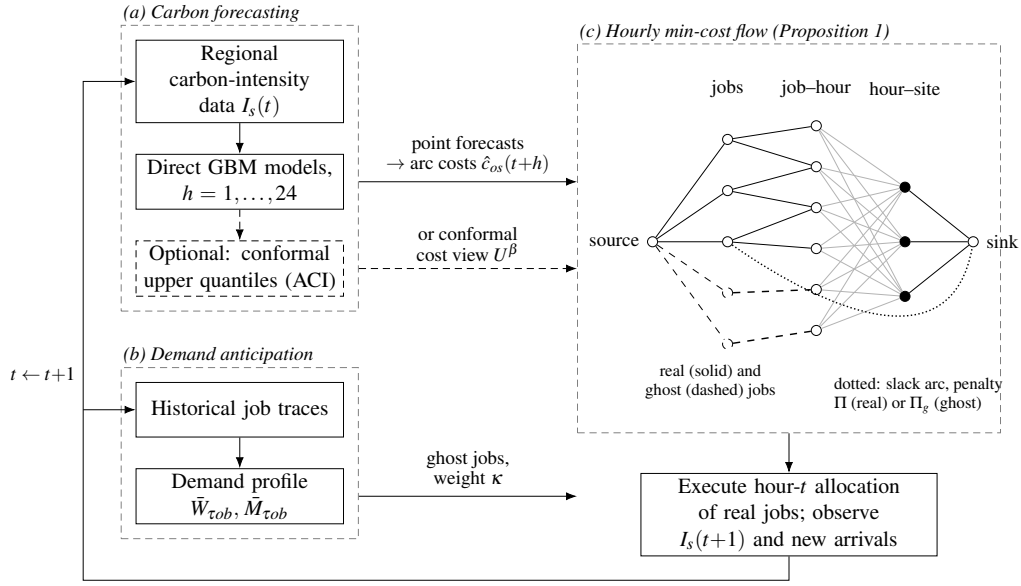


Figure 1: The CARMA decision loop. (a) Carbon-intensity forecasts price the job-hour to hour-site arcs; the selected configuration uses point forecasts, and conformal upper quantiles are an optional cost view. (b) A demand profile learned from past traces generates ghost jobs for expected arrivals. (c) Real jobs (solid) and ghost jobs (dashed) share one min-cost flow network, whose dotted slack arc carries unfinished work to the sink at the lateness penalty. The network is solved exactly every hour, and only the first-hour allocation of real jobs is executed.

326 *Carbon cost view.* Future-hour costs in Eq. (1) are evaluated at a forecast of
327 $I_s(t+h)$; for the current hour ($h=0$) the observed intensity is used. CARMA
328 admits two cost views: the point forecast $\hat{I}$, or the conformal upper quantile
329 U^β of Section 3.2. The second is motivated by the optimizer’s curse [55]:
330 selecting the minimum over noisy estimates is biased towards optimism. The
331 conformal view replaces each future cost by a calibrated quantile, which
332 shifts it by a site- and horizon-specific amount proportional to the local
333 volatility. For $\beta > 0.5$ this penalises volatile sites and distant hours more than
334 stable sites and near hours. Whether this risk premium helps is an empirical
335 question, and we treat the cost view as a hyperparameter.

336 *Algorithm.* Fig. 1 summarises the loop. At each hour t , CARMA (i) observes
337 $I_s(t)$, the free capacity and the pending jobs; (ii) builds the cost cube $c_{os}(t+h)$
338 for $h < H$ from the chosen forecast view; (iii) adds the ghost jobs for hours
339 $t+1, \dots, t+\ell$; (iv) solves the resulting min-cost flow problem exactly; (v)
340 executes the first-hour allocations of the real jobs only; and (vi) after observing
341 the realised intensities, updates the ACI levels by Eq. (7) when conformal
342 costs are used. Work that is still pending at its deadline receives a one-hour
343 planning window, which makes it urgent in the next plan; it carries the same
344 penalty as other real work, so no strict priority is imposed. The planning
345 network has $O((n_t + \ell|\mathcal{S}||\mathcal{B}|)H|\mathcal{S}|)$ arcs, where n_t is the number of pending
346 jobs, so each decision takes a few tens of milliseconds (Section 5). The
347 hyperparameters (κ, ℓ) and the cost view are tuned on validation weeks that
348 precede the test period (Section 5.2).

349 3.4. Baselines

350 We compare CARMA with the following online policies and the oracle.
351 **ASAP-Local** is carbon-agnostic: it runs work at its origin as soon as possible,
352 earliest deadline first (EDF). **Greedy-Spatial** also runs work immediately,
353 but at the site that is currently cleanest; this is geographical load balancing
354 without temporal shifting [13]. **Forecast-Reserve** books, when a job arrives,
355 the cheapest forecast (site, hour) slots in its window that still have capacity,
356 and never revises the booking. This is the lowest-window strategy of [10]
357 extended to multiple sites. **MPC** is the myopic receding-horizon flow scheduler
358 with point forecasts. **MPC-Protect** adds a protection level in the spirit
359 of revenue management: planned work may use only a share $1-r$ of the
360 capacity of the sites that are cleaner than the fleet median over the window.
361 In the current hour, the protected share is first offered to jobs due within τ

hours and then, because unused capacity perishes, to all remaining work. The parameters (r, τ) are tuned on validation data. **MPC+conformal** uses the conformal upper quantile at the level that was best for MPC on validation data. **MPC-Perfect** is MPC given the realised intensities, which isolates the value of forecast accuracy. **CARMA-Perfect** is CARMA given the realised intensities, and **CARMA+conformal** is CARMA with the best conformal cost view. **Scenario-MPC** is a demand-aware comparator that uses the same information as CARMA in a different way. At every hour it draws K arrival scenarios for the next ℓ hours from the historical job traces from which CARMA’s profile is estimated (the jobs of one historical week, or trace day, at the same hours), and solves a two-stage stochastic programme by sample-average approximation: the first-hour allocation of the pending jobs is shared, while the later allocation and the scenario jobs are planned separately in each scenario, and the mean cost is minimised. Unserved scenario work carries CARMA’s ghost penalty Π_g . The coupling across scenarios breaks the network structure, so Scenario-MPC is solved as an LP with HiGHS; (K, ℓ) are tuned on validation data. **Oracle** solves (2)–(5) with full knowledge of arrivals and intensities.

3.5. Data and experimental design

Carbon intensity. We use half-hourly regional carbon intensity for Great Britain from the Carbon Intensity API of the National Energy System Operator (NESO) [45] from 1 January 2022 to 31 December 2024. Values are averaged to hourly resolution. NESO does not meter regional intensity. For past periods, the regional endpoint returns a single modelled estimate per half-hour (in a field named `forecast`), and we treat this estimate as the realised intensity. The oracle and the “Perfect” variants are therefore clairvoyant with respect to NESO’s published regional values, and all emission figures inherit their modelling error. NESO’s forward regional forecasts are not archived, so they cannot serve as a historical benchmark for our forecaster. Four regions host the candidate sites: South Scotland, North West England, West Midlands and London. They span the north–south gradient of the British grid, from wind- and nuclear-rich South Scotland to gas-dependent London, and include the main data-centre markets of London, Manchester (North West England) and Birmingham (West Midlands). The largest cluster of the London market, around Slough, lies in NESO’s South England region; we use the London region as a stylised proxy for a southern, gas-dependent site. Our first download missed 602 of the 26,304 hours in at least one region,

399 most of them in two blocks of eleven to twelve days at the start of 2023 and
 400 2024. Re-requesting these days in one-day chunks recovered all but 52 hours,
 401 which the API does not provide: 41 hours on 20–22 October 2023 (in the
 402 tuning period) and 11 hours on 11–12 June 2024. These are filled causally by
 403 carrying the last observation forward, so no filled value uses later data. Filled
 404 hours are excluded from forecast training, calibration, ACI feedback and fore-
 405 cast evaluation. In the scheduling simulation they act as realised intensities;
 406 the two test weeks whose episodes contain them are checked separately in
 407 Section 5.3. The period up to 30 June 2023 is used to train the forecaster,
 408 July–September 2023 to calibrate the conformal scores, October–December
 409 2023 to tune the scheduler, and all 52 weeks starting on Mondays in 2024 as
 410 the out-of-sample test set.

411 *Sites and energy parameters.* Batch capacities are 300, 400, 400 and 600
 412 servers for South Scotland, North West England, West Midlands and London.
 413 This reflects a small, clean northern site and a large site in London; the split
 414 is varied in the sensitivity analysis. Interactive load takes $20\% \pm 15\%$ of each
 415 site’s capacity on a diurnal cycle that peaks in the afternoon. We assume
 416 $e = 0.4$ kWh per server-hour. The assumed PUE values are 1.15, 1.20, 1.25
 417 and 1.30 from north to south, reflecting free-cooling potential. Migrated work
 418 moves $\delta = 2$ GB per server-hour at $\eta = 0.02$ kWh/GB. This value lies within
 419 the wide range of published estimates of transmission energy intensity [62].
 420 The main experiments count only the energy of busy servers, which assumes
 421 that idle servers are switched off or reallocated. The sensitivity analysis varies
 422 η and the capacity split, and it adds an always-on case with an idle draw of
 423 0.2 kWh per server-hour for every batch server.

424 *Workloads.* Job arrivals follow a non-homogeneous Poisson process with a
 425 diurnal peak in the afternoon and 30% lower weekend rates. Jobs originate
 426 at the four sites with probabilities 0.10, 0.20, 0.25 and 0.45. Width m_j is
 427 log-uniform on $[4, 64]$ servers and the run time at full width r_j is uniform
 428 on $\{1, \dots, 6\}$ hours, so $w_j = m_j r_j$. A share v of jobs is urgent (deadline
 429 $= a_j + r_j$, no temporal slack); for the others the window is $\lceil f_j r_j \rceil$ hours with
 430 $f_j \sim U[1.5, 4]$, capped at 24 hours. The baseline value is $v = 0.3$. The arrival
 431 rate is scaled so that the offered load equals a fraction $\rho \in \{0.3, 0.5, 0.7\}$ of
 432 the mean free capacity of the fleet. Each test episode covers one week of
 433 arrivals. We also simulate 24 hours of arrivals before and after the week, so
 434 the evaluated week starts and ends under steady-state congestion. Metrics

435 are computed only for jobs released during the week. For each week and
 436 load level we draw three independent workload traces with fixed seeds, and
 437 every policy runs on the same traces. The demand profile used by CARMA is
 438 estimated from 13 independent historical traces of the same generator. This
 439 favours anticipation, because the profile is then an unbiased estimate of the
 440 arrival rate. We therefore also evaluate a trace-driven workload whose profile
 441 is learned from different days (next paragraph) and profiles learned at the
 442 wrong load (Section 5.5). We study two regimes: *spatio-temporal*, where
 443 migration is allowed, and *temporal-only*, where it is not. All hyperparameters
 444 are tuned separately for each regime.

445 *Trace-derived workload.* To test anticipation on real, non-stationary demand,
 446 we derive a fluid workload from the batch-task table of the Alibaba 2018
 447 cluster trace [63, 64]. The table has 14.3 million task records over eight
 448 days. We keep the 13.3 million terminated tasks with valid timestamps and a
 449 planned CPU request and group them by job (one directed acyclic graph of
 450 tasks), which gives 3.85 million jobs (script `prepare_trace.py`). The trace
 451 records no submission times, so a job’s arrival is the start of its first task. Its
 452 work is the sum over tasks of instances $\times$ run time $\times$ planned CPU request
 453 (core-hours), and its width is the largest planned core count of a single task;
 454 planned requests are not measured utilisation. Both are converted to servers
 455 of 96 cores and scaled so that the offered load in the evaluation days equals
 456 $\rho = 0.5$. Task precedence within a job is ignored. The trace records no
 457 deadlines, so windows are drawn as in the synthetic model: 30% of jobs are
 458 urgent, with a window equal to their run time rounded up to whole hours,
 459 and the others receive 1–23 hours of slack. Origins are assigned at random
 460 with the site probabilities above. To keep the planning problems tractable,
 461 jobs are aggregated into batches by arrival hour, origin and window class,
 462 and the works and widths of a batch are added. Pooling widths relaxes
 463 the per-job width limits, because a batch may run its short, wide members
 464 and its long, narrow members at a combined rate that no single member
 465 could sustain. The batched workload is therefore an approximation, and we
 466 test its effect with finer batches that also separate jobs by run-length class
 467 (Section 5.4). CARMA and Scenario-MPC learn from days 1–4 of the trace
 468 and are evaluated on days 5–8, preceded by 24 hours of warm-up. Days
 469 5–8 carry 39% more work than days 1–4, so the learned hour-of-day profile
 470 underestimates evaluation demand by 28%, and hourly work ranges from 0.2
 471 to 3.6 times its mean, so bursts are not anticipated. The four evaluation days

are paired with each of the 52 test weeks of 2024, starting on the Monday, with three independent draws of origins and deadlines. These 156 episodes vary the carbon conditions and the synthetic attributes, but they share one demand chronology.

Metrics and statistics. The primary metric is operational emissions (tCO₂) per week under average regional intensity, which is an *attributional* measure (Section 6 discusses marginal emissions). We report the reduction relative to ASAP-Local, the gap to the clairvoyant oracle, the share of late work, the share of work executed away from its origin and the decision time. Reductions and gaps are means of per-episode ratios, so they cannot be recomputed from the mean emissions in the tables. Late work is work still outstanding when its deadline passes; online policies keep processing it, all of it is completed before the end of the simulation, and its emissions are counted. The gap to the oracle is an optimality gap only for schedules that meet all deadlines. Where lateness is negligible, as in the spatio-temporal regime, it measures the loss of an online policy against the optimum. Where it is not, emissions and service must be read together, and we also report a service-adjusted gap. It charges each late server-hour a penalty equal to the largest local unit cost of the episode, in addition to its emissions, and compares with an oracle that minimises the same objective: evaluated work may be left undone at that penalty, while boundary work must still be served on time. The on-time part of any schedule that serves its boundary work on time is feasible for this oracle, so its value bounds the adjusted emissions of such schedules from below. The estimand for comparisons is the mean over the 52 test weeks of the paired difference in gap between a policy and CARMA, using week-level means over the three traces. Consecutive weeks are serially correlated (lag-1 autocorrelation of the paired differences ≈ 0.5), so confidence intervals and two-sided p -values come from a moving-block bootstrap over weeks [65] with blocks of four weeks and 5,000 draws. The p -value is the share of bootstrap means of the centred series that are at least as far from zero as the observed mean. We report block lengths of two and eight weeks as a sensitivity check. p -values are Holm-adjusted jointly over all comparisons of the main experiment (both regimes and all loads) and separately within the trace and sensitivity experiments.

506 4. Theoretical guarantees

507 This section establishes a worst-case limit for *every* online scheduler,
 508 prediction-dependent guarantees for CARMA, and a fairness property of the
 509 carbon attribution induced by the pooled scheduling problem. Proofs are
 510 in Appendix A. Throughout, unit costs (Eq. (1), including any backstop
 511 resource introduced below) satisfy $0 < L \leq c_{os}(t) \leq U$, and $\theta = U/L$ is
 512 the *unit-emission-cost ratio*. It refers to unit costs, which include energy
 513 use, PUE, transfer overheads and any backstop, not to grid intensities alone.
 514 Operational costs alone can be zero, because the hourly regional intensity
 515 of South Scotland is 0 gCO₂/kWh in 133 hours of our 2022–2024 data. A
 516 positive floor appears once the embodied emissions of the servers are allocated
 517 per server-hour, which is an accounting convention rather than a claim that
 518 scheduling avoids embodied emissions. As an illustration, a floor of 30 gCO₂e
 519 added to every unit cost, with the largest operational unit cost observed in
 520 2024 (221 g), gives $\theta = 251/30 \approx 8.4$. An online algorithm is Γ -*competitive*
 521 if its emissions never exceed Γ times those of the clairvoyant oracle, for any
 522 arrival sequence and any intensity trajectory.

523 4.1. A universal lower bound under demand uncertainty

524 Competitive analyses of carbon-aware scheduling model uncertainty about
 525 future *prices*, that is, intensities, for a single workload. Examples are online
 526 search and one-way trading [24], the online pause-and-resume problem [25],
 527 and spatiotemporal allocation with deadlines (CarbonClipper) [26]. When
 528 several jobs share scarce low-carbon capacity, difficulty remains even when
 529 all prices are known in advance, because future *demand* is unknown.

530 **Theorem 1** (Lower bound). *Let $\theta > 1$. No online algorithm (deterministic*
 531 *or randomised, with integral or fractional decisions) can be better than $R^*(\theta)$ -*
 532 *competitive, even if it knows all future carbon intensities exactly, where*

$$R^*(\theta) = \max_{1 \leq p \leq \theta} \left\{ 1 + \frac{(p-1)(\theta-p)}{p^2 - p - 1 + \theta} \right\} \geq 1 + \frac{\sqrt{\theta}(\sqrt{\theta}-1)^2}{2\theta - \sqrt{\theta} - 1} = \Omega(\sqrt{\theta}). \quad (8)$$

533 *On the same family of instances, the competitive ratio of myopic receding-*
 534 *horizon scheduling (MPC) is at least $(1+\theta)/2$, a value approached as $p \rightarrow 1$.*

535 For $\theta = 1$ every schedule costs the same and $R^*(1) = 1$; both expressions
 536 in (8) tend to 1 as $\theta \downarrow 1$.

537 The bound is established on a two-hour instance and is a quantitative
 538 version of Proposition 2, which treats the case $L = 0$. Any algorithm must
 539 decide how much flexible work to run now at moderate cost before it learns
 540 whether an urgent job will claim the cheap capacity later. The adversary
 541 thus raises the effective price of the clean hour from L to U . The resulting
 542 order, $\sqrt{\theta}$, is the same as for deterministic online min-search with unknown
 543 prices [24]. The difference is where it comes from: here every price is known,
 544 and the loss is caused only by the unknown arrival of a competing job. For
 545 the illustrative $\theta \approx 8.4$ above, $R^* = 1.81$, whereas myopic MPC can be driven
 546 to 4.7. For $\theta = 100$ the values are 5.29 and 50.5. $R^*(\theta)$ is a lower bound
 547 derived from one family of instances; we do not know whether any online
 548 algorithm attains it. Theorem 1 is a worst-case statement, but it points
 549 in the same direction as the average-case evidence of Section 5.3. Carbon
 550 forecasts cannot substitute for information about demand, and this motivates
 551 the learning-augmented design of CARMA [27].

552 4.2. *Deadline safety, consistency and smoothness of CARMA*

553 We analyse CARMA under two assumptions.

554 (A1) *Backstop capacity.* In every hour, work may also run on a backstop
 555 resource of unlimited capacity at a unit cost in $[L, U]$, for example
 556 by bursting to a public-cloud region or to reserve servers at a carbon-
 557 intensive site. The backstop’s unit cost is known in advance, or its
 558 forecast is clipped to $[L, U]$, so the planner never prices it above U . All
 559 jobs, real and predicted, are *window-feasible*, $w_j \leq m_j(d_j - a_j)$, and
 560 have windows of at most H hours, so every planning problem contains
 561 all pending deadlines (this holds by construction in our workloads). The
 562 penalties satisfy $\Pi > U$ for real jobs and $\Pi_g \geq U$ for ghost jobs. The
 563 implementation enforces window-feasible ghost jobs (Section 3.3) and
 564 sets Π_g to 1.5 times the largest local unit cost, which exceeds every unit
 565 cost including transfer overheads.

566 (A2) *Episode within the planning window.* The episode $\{0, \dots, T - 1\}$ has
 567 length $T \leq H$, as for a daily batch window, so every planning problem
 568 extends to the end of the episode.

569 To measure prediction quality at hour t , let F_t be the set of jobs that actually
 570 arrive after t , and $\hat{F}_t$ the ghost jobs used by CARMA. Jobs are compared
 571 after splitting them into *atoms* with proportional widths, which leaves every

572 planning problem unchanged (Appendix A). Let d_t be the smallest total
 573 work of atoms that must be added to or removed from $\hat{F}_t$ to obtain F_t . Let
 574 $\varepsilon_t = \max_{\tau > t, o, s} |\hat{c}_{os}(\tau | t) - c_{os}(\tau)|$ be the carbon-cost forecast error, and W_t
 575 the total work pending at t plus the larger of the true and predicted future
 576 work.

577 **Proposition 3** (Baseline robustness). *Under (A1), every policy that completes*
 578 *all work by its deadlines is θ -competitive.*

579 **Theorem 2** (Guarantees for CARMA). *Under (A1), for any arrival sequence*
 580 *and any predictions, CARMA completes every job by its deadline (deadline*
 581 *safety) and is therefore θ -competitive. If (A2) also holds, then*

$$\text{ALG} \leq \min \left\{ \theta \cdot \text{OPT}, \text{OPT} + \sum_{t=0}^{T-1} [(U - L) d_t + 2 \varepsilon_t W_t] \right\}. \quad (9)$$

582 *In particular, CARMA is optimal with exact predictions ($d_t = \varepsilon_t = 0$), which*
 583 *is consistency, and its excess emissions grow at most linearly in the prediction*
 584 *errors, which is smoothness.*

585 These properties should be read with care. Robustness by itself does not
 586 distinguish CARMA: by Proposition 3 it holds for any policy that meets
 587 its deadlines. What CARMA adds is deadline safety under *arbitrary* ghost
 588 predictions, together with consistency and smoothness. Deadline safety holds
 589 because ghost jobs carry a softer penalty than real jobs and the backstop can
 590 always absorb real work, so a misprediction can divert real work to dirtier
 591 capacity but never make it late. Without a backstop this is no longer true.
 592 With one server and two hours, a flexible job that could run in the first hour
 593 may be deferred to a cleaner second hour, and an unexpected urgent job in
 594 the second hour then makes one of them late, although an offline schedule
 595 meets both deadlines. A soft ghost penalty does not create capacity. “Exact
 596 predictions” in the theorem means exact predictions of both demand and
 597 carbon; the CARMA-Perfect variant of the experiments has perfect carbon
 598 forecasts but still uses the average demand profile and a finite look-ahead,
 599 so it is not the case $d_t = 0$. The smoothness term charges demand errors
 600 at the spread $U - L$ of unit costs, not at the lateness penalty. The term is
 601 informative when predictions are close to the realised demand. For random
 602 arrivals, however, an expected-value profile never matches the realised jobs
 603 exactly. Moreover, each misprediction enters d_t in every epoch in which it is

604 visible, so the bound in (9) can be loose. In the main experiments, assumption
 605 (A2) is not satisfied (weekly episodes with $H = 24$ h), no backstop is modelled,
 606 and some hours have zero operational cost. The theorem therefore describes
 607 an idealised setting and does not certify the reported numbers; the degradation
 608 profile in the weekly experiments is tested empirically in Section 5.6. Appendix
 609 C checks the theorem numerically in one-day episodes that do satisfy (A1) and
 610 (A2). Robustness θ is also far from the $\Omega(\sqrt{\theta})$ lower bound. Closing this gap
 611 while retaining consistency is left for future work, for example by protecting
 612 deferral decisions with reservation prices [24, 26]. Fig. 2 summarises the
 613 worst-case landscape.

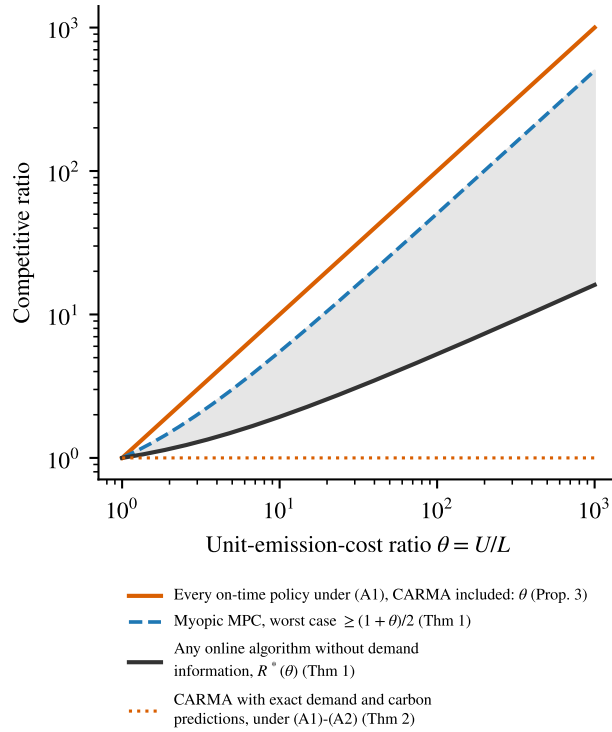


Figure 2: Worst-case landscape as a function of the unit-emission-cost ratio $\theta = U/L$. The curves refer to different information regimes. Without information about future demand, no online algorithm can lie below $R^*(\theta)$, even with exact carbon forecasts (Theorem 1), and myopic MPC can be driven to $(1 + \theta)/2$. Given a backstop (A1), every policy that meets its deadlines, CARMA included, is within θ (Proposition 3). With exact predictions of *both* demand and carbon and an episode within the planning window (A1–A2), CARMA is optimal (Theorem 2).

614 4.3. Fair carbon attribution among sites

615 Pooling capacity reduces total emissions but redistributes them. Work
 616 submitted in London is executed in South Scotland, and Scottish jobs may be
 617 displaced by it. The organisations that operate the sites, or the business units
 618 that submit the jobs, therefore need an attribution of the pooled emissions
 619 that each of them can accept. This is an *internal* allocation problem, like
 620 carbon transfer pricing between units. It is distinct from external GHG
 621 Protocol reporting, where Scope 2 emissions are reported by the entity that
 622 consumes the electricity [66].

623 We model the allocation as a cooperative game. The players are the sites $\mathcal{S}$.
 624 A coalition $C \subseteq \mathcal{S}$ pools the capacity of its sites and serves the jobs submitted
 625 at them. Its stand-alone value $c(C)$ is the optimum of (2)–(5) restricted
 626 to those jobs and sites, which includes the lateness penalty if the coalition
 627 cannot serve its own jobs on time. An attribution $\phi \in \mathbb{R}^{|\mathcal{S}|}$ is *budget-balanced*
 628 if $\sum_o \phi_o = c(\mathcal{S})$. It is in the *core* if $\sum_{o \in C} \phi_o \leq c(C)$ for every coalition C . A
 629 core attribution gives no group of sites a reason to leave the pool and schedule
 630 on its own.

631 Let (λ, ν, μ) be optimal dual variables of the grand-coalition problem: λ_j
 632 for the work constraint (3), $\nu_{jt} \leq 0$ for the width constraint (4) and $\mu_{st} \leq 0$
 633 for the capacity constraint (5). The *dual carbon attribution* is

$$\phi_o = \sum_{j: o_j=o} \left(\lambda_j w_j + \sum_t \nu_{jt} m_j \right) + \sum_t \mu_{ot} K_o(t). \quad (10)$$

634 Site o is charged the carbon shadow price of the work it submits and is
 635 credited ($\mu \leq 0$) with the shadow value of the capacity it contributes. A
 636 site whose clean capacity is worth more than the emissions of its own jobs
 637 therefore receives a net credit, that is, a negative internal charge.

638 **Theorem 3** (Core-stable carbon attribution). *The attribution (10) is budget-*
 639 *balanced and in the core of the pooling game. In particular, it is individually*
 640 *rational: no site is charged more than its stand-alone optimum $c(\{o\})$.*

641 The result follows from Owen’s theorem on linear production games [31];
 642 games defined by network flows are known to be totally balanced [32]. It
 643 concerns the penalised hindsight game defined above. Core stability is one
 644 requirement of an acceptable allocation; it does not imply uniqueness, incentive
 645 compatibility or other notions of fairness, and it does not extend automatically
 646 to the realised emissions of an online policy. The duals are those of the

hindsight problem, which suits after-the-fact accounting. They are available as the optimal node potentials of the min-cost flow; in our experiments we obtain them from an LP solver. Optimal duals need not be unique, so Section 5.7 reports the range of (10) over all optimal duals. It also compares the dual attribution with the Shapley value [67], which is budget-balanced but not guaranteed to lie in the core, and with simpler rules. The simpler rules are *origin-based* (each site is charged the emissions of its own jobs wherever they ran), *host-based* (each site is charged the emissions of the work it executes, the analogue of location-based Scope 2 reporting), and proportional rules.

5. Results

5.1. Carbon-intensity data and forecasts

Fig. 3 shows the regional intensities. Over 2022–2024, mean intensity was 37 gCO₂/kWh in South Scotland, 58 in North West England, 156 in the West Midlands and 163 in London. The regional means indicate large spatial heterogeneity, and it persists hour by hour: the spread between the cleanest and the dirtiest of the four sites has a median of 156 gCO₂/kWh and exceeds 54 g in 90% of hours. This is what makes spatial shifting attractive. Annual mean intensities declined at every site; London’s fell from 202 gCO₂/kWh in 2022 to 125 in 2024. Diurnal profiles peak in the early evening (around 18:00 UTC), and the profile of South Scotland is much flatter than those of the southern sites. The four series are positively correlated (pairwise correlations between 0.51 and 0.87), because national wind output drives all of them.

The forecaster was trained on 1.24 million (issue time, site, horizon) samples in about one minute. In 2024 its mean absolute error (MAE), averaged over horizons and sites, is 31.7 gCO₂/kWh, against 35.3 for persistence and 40.9 for a seasonal-naive forecast. The advantage is 23–25% at horizons of 1–6 hours, and the difference is significant at all English sites up to 18 hours in Diebold–Mariano tests [68]. Day-ahead skill is limited, because no weather inputs are used. Adaptive recalibration keeps the annual coverage of the upper bounds within 0.003 of target despite the 2024 decarbonisation drift. Appendix B gives the full evaluation (accuracy by horizon and site, coverage, pinball loss).

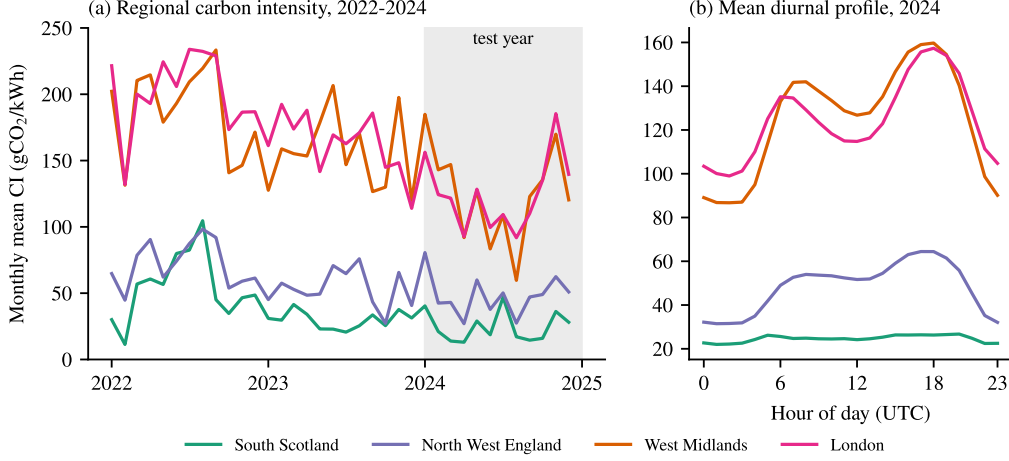


Figure 3: Hourly carbon intensity of the four candidate sites (NESO regional data). (a) Monthly means, 2022–2024; the shaded area is the out-of-sample test year. (b) Mean diurnal profile in 2024 (UTC).

679 5.2. Hyperparameter tuning

680 All policies were tuned on 13 validation weeks (October–December 2023,
681 $\rho \in \{0.3, 0.5, 0.7\}$), separately for each regime. Fig. 4 shows the spatio-
682 temporal regime. Anticipating demand helps across the whole grid: every
683 tested configuration beats myopic MPC, whose mean gap to the oracle is 22.3%.
684 The gap is smallest around $\kappa = 1$, where expected demand is anticipated
685 without inflation: 9.3% for $\ell = 12$ and 9.1% for $\ell = 18$. Over-anticipation
686 ($\kappa = 1.25$ with $\ell = 18$) reserves too much clean capacity and is penalised
687 (12.4%). We selected $\kappa = 1$ and $\ell = 12$ hours, resolving configurations within
688 0.25 percentage points of the best towards the shorter look-ahead. In the
689 temporal-only regime, anticipation brings no benefit. The best configuration
690 ($\kappa = 0.5$, $\ell = 6$; 4.29%) is indistinguishable from MPC (4.29%), and larger κ
691 is slightly worse. For MPC-Protect, the best protection level was $r = 0.3$ with
692 $\tau = 1$ h in the spatio-temporal regime (21.3% against 22.3% for MPC). In the
693 temporal-only regime every protection level was worse than MPC (at best
694 5.9%). For Scenario-MPC, $K = 8$ scenarios with $\ell = 12$ gave 9.0%, against
695 9.3% with $K = 4$ and 12.5–12.7% with $\ell = 6$.

696 The cost view interacts with anticipation (Fig. 4b). For myopic MPC,
697 high conformal levels help. The gap falls from 25.5% at $\beta = 0.5$ to 17.3%
698 at $\beta = 0.9$ and rises again at $\beta = 0.95$ (18.5%), against 22.3% with point

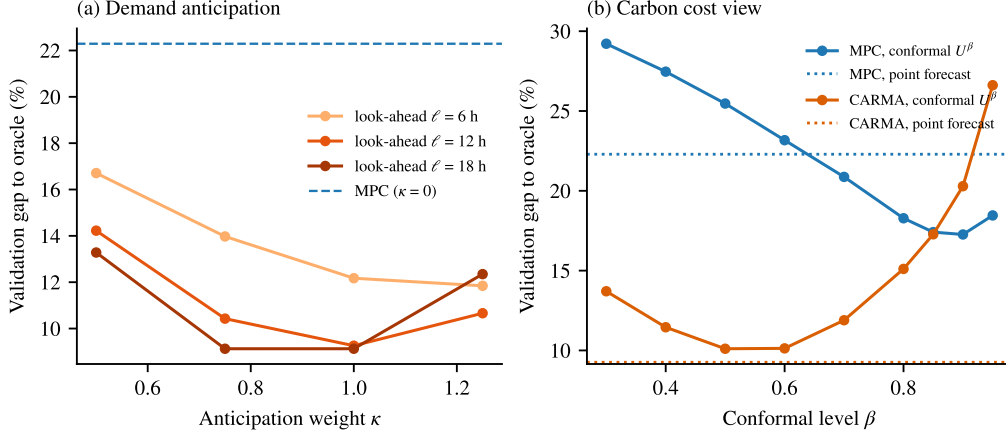


Figure 4: Validation results for the spatio-temporal regime (2023-Q4, mean over 13 weeks and three load levels). (a) Gap to the oracle as a function of the anticipation weight κ and look-ahead ℓ , using point forecasts. (b) Effect of the carbon cost view: conformal quantiles U^β against point forecasts, for myopic MPC and for CARMA ($\kappa = 1$, $\ell = 12$).

forecasts. For CARMA, every conformal level is worse than the point forecast: the best, $\beta = 0.5$, gives 10.1% against 9.3%. The mechanism is not a uniform risk premium. On the validation weeks, the $\beta = 0.6$ view lies on average between 1.5 g below and 3.9 g above the point forecast at South Scotland, North West England and London, but 14.5 g above it at the volatile West Midlands site. The offset also grows with the horizon, and at $\beta = 0.8$ it reaches 23–95 g at a 24-hour horizon. High levels therefore make distant hours and volatile sites look dirtier, so the planner runs work now and at stable sites, which keeps future clean capacity free. For a myopic planner this is a crude form of anticipation. Once CARMA represents future demand explicitly, the same distortion double-counts it and gives up useful deferrals. The selected CARMA therefore uses point forecasts. For the test we keep MPC+conformal at $\beta = 0.9$ and CARMA+conformal at $\beta = 0.5$ (spatio-temporal) as ablations. In the temporal-only regime, the best conformal views were $\beta = 0.6$ for MPC and $\beta = 0.5$ for CARMA.

5.3. Out-of-sample scheduling performance

Tables 2 and 3 report results for the 52 test weeks of 2024, with three independent workload traces per week. Weekly traces contain on average 917, 1,527 and 2,141 jobs (69,400, 115,400 and 162,000 server-hours) at $\rho = 0.3$,

0.5 and 0.7. Fig. 5 shows the gaps to the oracle.

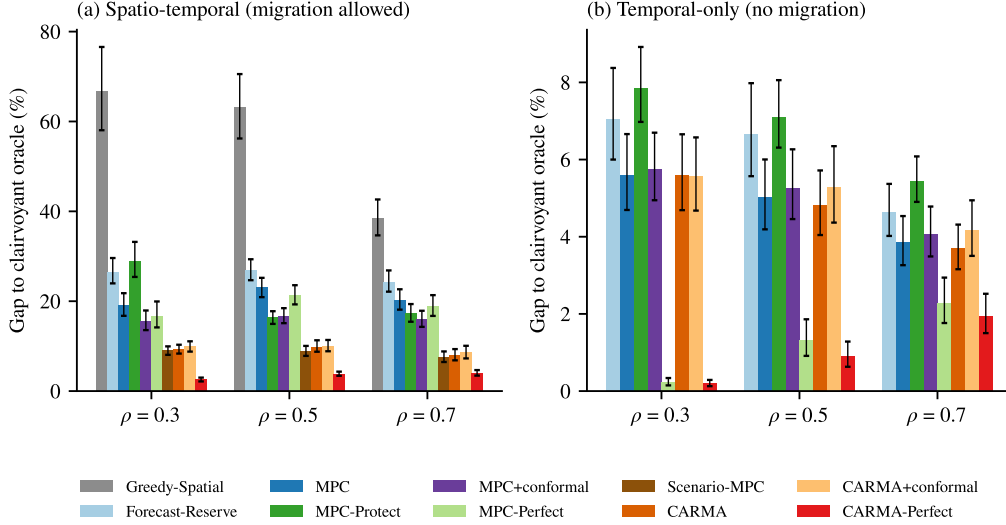


Figure 5: Mean gap to the clairvoyant oracle over the 52 test weeks of 2024 (week means of three traces) with 95% confidence intervals. ASAP-Local is omitted (gap 72–280% with migration, 9–22% without).

Spatio-temporal regime. With migration allowed, the oracle cuts attributional emissions relative to ASAP-Local by 71.8%, 58.2% and 41.1% at $\rho = 0.3$, 0.5 and 0.7. Greedy-Spatial realises only part of this potential (gaps of 38–67%), because it has no temporal flexibility and floods the cleanest site with whatever arrives first. The forecast-driven planners without demand information come closer. Forecast-Reserve has gaps of 24–27% and MPC 19–23%. MPC+conformal (15.5–16.6%) is the best of them at $\rho = 0.3$ and 0.7 and MPC-Protect at $\rho = 0.5$ (16.3%); at $\rho = 0.3$, where clean capacity is rarely scarce, the protection level wastes it and MPC-Protect is worse than MPC (28.8%). CARMA reduces emissions by 69.3%, 54.3% and 36.6% and has gaps of 9.3%, 9.8% and 8.0%. Relative to MPC this is a further emission reduction of 7.7%, 9.7% and 9.1%. Its advantage over the best of these baselines at each load is 6.3 (95% block-bootstrap CI 4.5–8.2), 6.5 (5.2–7.4) and 8.0 (6.9–9.0) percentage points of gap, and CARMA emits less than MPC in 51, 51 and 52 of the 52 week-means. Scenario-MPC, which uses the same demand information in a two-stage stochastic programme, does

Table 2: Out-of-sample results in the spatio-temporal regime (migration allowed), 52 weeks of 2024×3 traces. Emis.: mean weekly operational emissions (tCO₂, attributional). Red.: reduction relative to ASAP-Local (%). Gap: excess over the clairvoyant oracle (%). Adj.: service-adjusted gap, in which late work is additionally charged at the episode’s largest local unit cost and the oracle minimises the same objective. Late: share of work outstanding at its deadline (%). Mig.: share of work executed away from its origin (%). ms: mean decision time per hour (one CPU core). p : two-sided moving-block bootstrap test of the mean paired weekly difference in gap against CARMA, Holm-adjusted jointly over all comparisons in Tables 2 and 3. Red. and Gap are means of per-episode ratios. The best gap among implementable online policies is in bold.

Load	Policy	Emis.	Red.	Gap	Adj.	Late	Mig.	ms	p
$\rho = 0.3$	ASAP-Local	3.71	0.0	284.0	284.0	0.00	0.0	0.03	<0.001
	Greedy-Spatial	1.72	54.7	66.8	66.8	0.00	81.8	0.10	<0.001
	Forecast-Reserve	1.34	64.9	26.4	26.4	0.00	83.8	0.39	<0.001
	MPC	1.28	66.8	19.1	19.1	0.00	69.5	2.1	<0.001
	MPC-Protect	1.37	64.2	28.8	28.8	0.00	80.5	1.8	<0.001
	MPC+conformal	1.24	67.7	15.5	15.5	0.00	70.0	1.9	<0.001
	MPC-Perfect	1.25	67.5	16.7	16.7	0.00	69.1	2.1	<0.001
	Scenario-MPC	1.18	69.4	9.0	9.0	0.00	70.2	248	0.019
	CARMA	1.18	69.3	9.3	9.3	0.00	70.1	21	–
	CARMA+conformal	1.18	69.2	9.9	9.9	0.00	70.3	21	0.185
	CARMA-Perfect	1.12	71.1	2.5	2.5	0.00	69.9	21	<0.001
	Oracle	1.09	71.8	0.0	0.0	0.00	70.9	–	–
$\rho = 0.5$	ASAP-Local	6.11	0.0	149.6	150.0	0.05	0.0	0.05	<0.001
	Greedy-Spatial	4.11	33.6	63.1	63.1	0.00	76.5	0.17	<0.001
	Forecast-Reserve	3.29	47.6	26.8	27.0	0.02	78.3	0.65	<0.001
	MPC	3.19	49.2	23.0	23.1	0.01	50.1	4.2	<0.001
	MPC-Protect	3.05	51.7	16.3	16.3	0.00	58.8	4.9	<0.001
	MPC+conformal	3.05	51.6	16.6	16.6	0.00	50.5	4.0	<0.001
	MPC-Perfect	3.15	49.9	21.3	21.4	0.01	49.3	4.1	<0.001
	Scenario-MPC	2.87	54.7	8.8	8.8	0.00	50.7	674	<0.001
	CARMA	2.88	54.3	9.8	9.8	0.00	50.3	31	–
	CARMA+conformal	2.89	54.3	9.9	10.0	0.00	50.6	31	1.000
	CARMA-Perfect	2.75	56.7	3.8	3.8	0.00	50.1	31	<0.001
	Oracle	2.66	58.2	0.0	0.0	0.00	51.8	–	–
$\rho = 0.7$	ASAP-Local	8.31	0.0	72.6	99.6	4.54	0.0	0.08	<0.001
	Greedy-Spatial	6.76	19.3	38.4	38.5	0.00	73.3	0.26	<0.001
	Forecast-Reserve	6.12	27.3	24.3	30.8	1.12	73.6	1.1	<0.001
	MPC	5.92	29.7	20.2	31.2	1.90	31.6	6.6	<0.001
	MPC-Protect	5.79	31.3	17.2	21.1	0.69	37.8	8.1	<0.001
	MPC+conformal	5.74	32.1	16.0	18.1	0.36	32.1	6.5	<0.001
	MPC-Perfect	5.85	30.5	18.8	30.5	1.96	31.4	6.3	<0.001
	Scenario-MPC	5.36	36.8	7.5	7.6	0.01	31.2	1453	<0.001
	CARMA	5.38	36.6	8.0	8.0	0.02	31.5	40	–
	CARMA+conformal	5.41	36.2	8.5	8.8	0.05	31.5	40	0.034
	CARMA-Perfect	5.21	38.7	4.0	4.2	0.04	31.0	40	<0.001
	Oracle	5.04	41.1	0.0	0.0	0.00	32.7	–	–

Table 3: Out-of-sample results in the temporal-only regime (no migration), 52 weeks of 2024 $\times$ 3 traces. Columns as in Table 2. Greedy-Spatial is omitted because without migration it coincides with ASAP-Local.

Load	Policy	Emis.	Red.	Gap	Adj.	Late	ms	p
$\rho = 0.3$	ASAP-Local	3.71	0.0	22.3	22.3	0.00	0.03	<0.001
	Forecast-Reserve	3.27	12.3	7.1	8.6	0.38	0.33	<0.001
	MPC	3.23	13.4	5.6	5.6	0.00	1.3	1.000
	MPC-Protect	3.30	11.6	7.8	7.8	0.00	1.6	<0.001
	MPC+conformal	3.24	13.3	5.7	5.7	0.00	1.3	1.000
	MPC-Perfect	3.09	17.8	0.2	0.3	0.01	1.4	<0.001
	CARMA	3.23	13.4	5.6	5.6	0.00	6.9	–
	CARMA+conformal	3.23	13.5	5.6	5.6	0.01	7.2	1.000
	CARMA-Perfect	3.08	17.8	0.2	0.2	0.01	7.3	<0.001
	Oracle	3.08	18.0	0.0	0.0	0.00	–	–
$\rho = 0.5$	ASAP-Local	6.11	0.0	18.0	18.2	0.05	0.05	<0.001
	Forecast-Reserve	5.54	9.5	6.6	18.3	3.03	0.66	<0.001
	MPC	5.47	10.9	5.0	6.8	0.52	2.5	<0.001
	MPC-Protect	5.58	9.1	7.1	8.7	0.48	2.9	<0.001
	MPC+conformal	5.48	10.7	5.3	6.9	0.47	2.5	<0.001
	MPC-Perfect	5.29	14.0	1.3	4.2	0.78	2.6	<0.001
	CARMA	5.46	11.0	4.8	6.0	0.36	9.7	–
	CARMA+conformal	5.48	10.7	5.3	7.2	0.54	9.8	0.006
	CARMA-Perfect	5.28	14.3	0.9	2.9	0.54	10	<0.001
	Oracle	5.23	15.1	0.0	0.0	0.01	–	–
$\rho = 0.7$	ASAP-Local	8.31	0.0	9.6	22.5	4.54	0.07	<0.001
	Forecast-Reserve	7.93	4.5	4.6	30.7	8.15	1.2	<0.001
	MPC	7.88	5.2	3.8	42.1	11.80	4.5	0.006
	MPC-Protect	8.00	3.8	5.4	42.8	11.54	5.1	<0.001
	MPC+conformal	7.90	5.0	4.1	42.0	11.54	4.5	<0.001
	MPC-Perfect	7.77	6.6	2.3	50.0	14.03	4.6	<0.001
	CARMA	7.87	5.3	3.7	32.9	9.26	14	–
	CARMA+conformal	7.90	4.9	4.2	40.6	11.09	14	<0.001
	CARMA-Perfect	7.75	6.9	1.9	41.9	11.85	14	<0.001
	Oracle	7.62	8.7	0.0	0.0	0.76	–	–

735 slightly better than CARMA: its gaps are 9.0%, 8.8% and 7.5%, lower by
736 0.3 (0.2–0.5), 1.0 (0.9–1.3) and 0.4 (0.3–0.6) points, and CARMA is better
737 in only 19, 5 and 8 weeks. Scenario-MPC, however, needs 250–1,450 ms per
738 decision against 21–40 ms for CARMA, 12–37 times longer, and solves a
739 general LP instead of a network flow. CARMA thus captures most of the
740 value of demand information, 10–13 points of gap against MPC, and gives
741 up 0.3–1.0 points against a stochastic programme with the same information.
742 Block lengths of two and eight weeks give the same conclusions.

743 CARMA also finishes more work on time under heavy load. At $\rho = 0.7$,
744 0.02% of its work is late (Scenario-MPC 0.02%), against 1.9% for MPC, 0.7%
745 for MPC-Protect, 1.1% for Forecast-Reserve and 4.5% for ASAP-Local. The
746 service-adjusted gap therefore widens the difference: 8.0% for CARMA against
747 18.1–31.2% for the forecast-driven baselines without demand information.
748 CARMA migrates about the same share of work as MPC (32–70%), whereas
749 MPC-Protect migrates more (38–81%). CARMA’s advantage thus comes
750 from better timing of the use of scarce clean capacity, not from moving more
751 data. For the fleet of 1,700 batch servers at $\rho = 0.5$, CARMA emits 0.31 tCO₂
752 per week less than MPC, about 16 tCO₂ per year, and 3.22 tCO₂ per week
753 less than carbon-agnostic operation. These absolute figures scale with fleet
754 size. Two test weeks contain gap-filled hours (Section 3.5); excluding them
755 changes CARMA’s gaps by at most 0.2 points and no ranking.

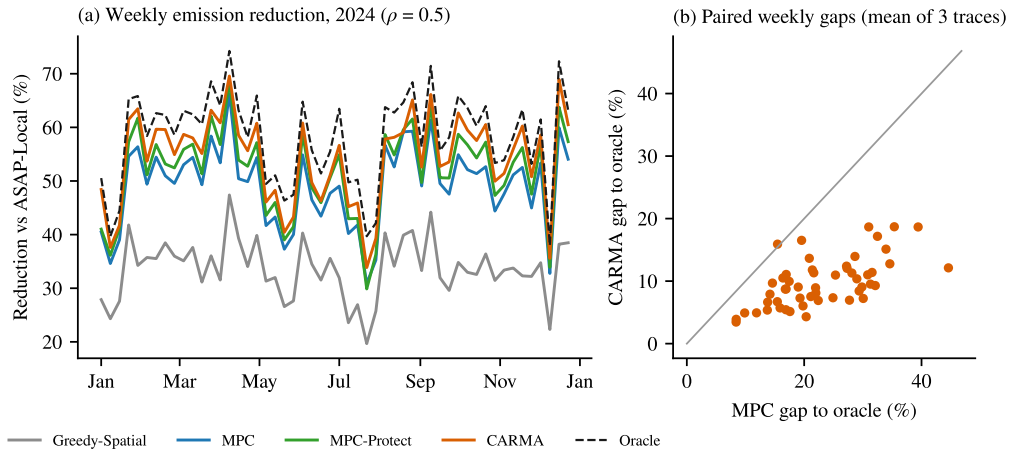


Figure 6: Weekly results in 2024 (spatio-temporal regime, $\rho = 0.5$, means of three traces). (a) Emission reduction relative to ASAP-Local. (b) Paired weekly gaps to the oracle; points below the diagonal are weeks in which CARMA outperforms MPC.

756 *Value of information.* The clairvoyant variants separate two sources of loss.
757 With perfect intensity forecasts, the myopic MPC-Perfect still has gaps of
758 16.7%, 21.3% and 18.8%. Perfect carbon foresight is thus worth only 1.4–2.4
759 percentage points to a myopic planner, whereas anticipating demand is worth
760 9.8–13.2 points (MPC versus CARMA). With both, CARMA-Perfect comes
761 within 2.5–4.0% of the oracle. CARMA-Perfect has exact carbon forecasts
762 but still uses the average demand profile, so it is not the exact-prediction
763 case of Theorem 2. The two kinds of information are complementary. Once
764 demand is anticipated, forecast accuracy is worth 4.0–6.8 points (CARMA
765 versus CARMA-Perfect), about three to four times more than for the myopic
766 planner. The ranking is stable across seasons (Fig. 6a), although the level of
767 achievable reductions varies from week to week with wind conditions.

768 *Temporal-only regime.* Without migration (Table 3), each job can only move
769 in time at its origin. Achievable reductions are much smaller: the oracle
770 saves 18.0%, 15.1% and 8.7%. MPC and CARMA are within 3.7–5.6% of
771 the oracle. They are statistically indistinguishable at $\rho = 0.3$ (difference 0.01
772 points, CI -0.01 to 0.02). CARMA is slightly better at higher loads, by 0.20
773 (0.13–0.30) and 0.15 (0.07–0.25) points. Both are better than Forecast-Reserve
774 and MPC-Protect. Anticipation brings little benefit here, because each site’s
775 capacity serves only its own jobs, and flexible and urgent work rarely compete
776 for the same clean hours in a way that deferral cannot absorb. The remaining
777 loss is due almost entirely to forecast error: the Perfect variants close the
778 gap to 0.2–2.3%. Heavy load exposes a trade-off that the emission metric
779 hides. At $\rho = 0.7$, the capacity at the origin sites is itself insufficient, and
780 every carbon-aware policy delays work: 9.3% of CARMA’s work and 11.8%
781 of MPC’s is late, against 4.5% for ASAP-Local. In this cell the emission
782 gap is not an optimality gap, because the oracle meets deadlines. Under the
783 service-adjusted metric, carbon-agnostic operation (22.5%) is preferable to
784 MPC (42.1%) and CARMA (32.9%). Without migration and near saturation,
785 temporal shifting should therefore be restricted or combined with explicit
786 service constraints.

787 *Computation.* A CARMA decision takes 7–40 ms on one CPU core, against
788 1–7 ms for MPC; the difference comes from the ghost jobs, which enlarge
789 the planning network. Scenario-MPC needs 0.25–1.5 s. Both are fast enough
790 for hourly decisions in this fleet, but the cost of the stochastic programme
791 grows with the number of scenarios and jobs, while CARMA remains a single

792 network flow. A clairvoyant weekly oracle with up to about 2,800 jobs is
 793 solved in under one second by the flow solver.

794 5.4. *Trace-derived workload*

795 Table 4 reports the experiment with the trace-derived workload. Arrivals
 796 and job sizes come from the Alibaba trace, and the demand information
 797 of CARMA and Scenario-MPC comes from different days, so the profile
 798 underestimates demand by 28% and cannot anticipate bursts. The evaluation
 799 covers the four trace days paired with 52 carbon windows of 2024, with three
 800 draws of origins and deadlines each, and 3,072 evaluated job batches (65,300
 801 server-hours) per window. Real, bursty demand makes all policies worse than
 802 in the synthetic experiment, but the ranking is unchanged. CARMA has a gap
 803 of 21.2%, against 27.5% for MPC-Protect, the best baseline without demand
 804 information, and 36.8% for MPC. Carbon foresight again matters less than
 805 demand anticipation: MPC-Perfect reaches only 34.1%. CARMA’s advantage
 806 over MPC-Protect is 6.3 percentage points (95% CI 4.6–8.3; CARMA better
 807 in 42 of 52 carbon windows, $p < 0.001$). Scenario-MPC is again slightly
 808 better than CARMA, by 1.4 points (0.6–1.8; gap 19.7%), at 1.2 s per decision
 809 against 48 ms. The bias of the learned profile costs 3.0 points (CI 1.1–4.0):
 810 CARMA with a profile estimated in-sample from the evaluation days reaches
 811 18.2%. With the biased profile but perfect carbon forecasts, the gap would
 812 fall by 3.8 points (CARMA-Perfect, 17.4%). The remaining distance to the
 813 oracle reflects bursts that no profile anticipates. CARMA also keeps late work
 814 low (0.09%, against 0.66% for MPC).

815 Batching pools the widths of jobs with different work-to-width ratios
 816 and therefore relaxes the problem. With finer batches that also separate
 817 jobs by run-length class (about twice as many batches, one draw per carbon
 818 window), the oracle emits 2.5% more, so batching makes the benchmark
 819 slightly optimistic. The ranking is unchanged: gaps are 20.0% for CARMA,
 820 26.2% for MPC-Protect and 34.3% for MPC, and CARMA’s advantage over
 821 MPC-Protect is 6.3 points (4.8–7.8). The 156 episodes share one demand
 822 chronology, so the trace result is evidence about robustness to carbon condi-
 823 tions and to a biased profile for this workload, not a sample of independent
 824 demand weeks.

Table 4: Trace-derived workload (Alibaba 2018 cluster trace, days 5–8; demand information from days 1–4), paired with 52 carbon windows of 2024×3 draws of origins and deadlines; spatio-temporal regime, $\rho = 0.5$. Emissions are per four-day window. Columns as in Table 2; the service-adjusted gap is omitted; p values are Holm-adjusted within this table.

Policy	Emis.	Red.	Gap	Late	Mig.	ms	p
ASAP-Local	3.67	0.0	198.8	0.00	0.0	0.06	<0.001
Greedy-Spatial	2.48	33.4	94.2	0.00	75.0	0.19	<0.001
Forecast-Reserve	1.92	49.4	43.4	0.82	77.5	1.8	<0.001
MPC	1.84	51.7	36.8	0.66	52.9	16	<0.001
MPC-Protect	1.73	54.6	27.5	0.06	63.9	16	<0.001
MPC-Perfect	1.80	52.7	34.1	0.63	52.3	15	<0.001
Scenario-MPC	1.65	57.1	19.7	0.07	53.3	1160	<0.001
CARMA	1.66	56.7	21.2	0.09	53.4	48	–
CARMA-Perfect	1.61	58.1	17.4	0.15	51.9	49	<0.001
Oracle	1.40	63.8	0.0	0.00	56.5	–	–
CARMA (in-sample profile)	1.62	57.6	18.2	0.00	52.2	48	<0.001

825 5.5. Sensitivity analysis

826 Table 5 varies one factor at a time around the base case: spatio-temporal
827 regime, $\rho = 0.5$, $v = 0.3$, $\eta = 0.02$ kWh/GB, capacities 300/400/400/600.
828 Each setting uses 26 alternating test weeks of 2024 with two fresh workload
829 traces per week. All policies keep their base-case hyperparameters.

830 *Workload and network.* CARMA has the smallest gap of the online policies
831 compared here in every setting (Scenario-MPC is omitted from the sensitivity
832 analysis because of its computational cost). Except under marginal accounting
833 (discussed below), it emits less than MPC in all 26 weeks of each setting.
834 The benefit of anticipation grows with the share of urgent work, because
835 urgent arrivals are precisely the demand that myopic planners crowd out.
836 With $v = 0.5$, CARMA’s gap is 6.7% against 15.3% for MPC-Protect and
837 21.5% for MPC. With $v = 0.1$, it is 12.0% against 16.4% and 23.3%. The
838 ranking is unchanged over a twelve-fold range of transmission energy intensity
839 ($\eta = 0.005$ – 0.06 kWh/GB), and the migrated share of CARMA’s work stays
840 at 49–51%.

841 *Capacity layout.* Arrival rates are rescaled with the fleet’s capacity, so the
842 offered load stays at $\rho = 0.5$ of the new capacity. The advantage does not
843 depend on the chosen layout. When the clean South Scotland site is halved
844 (150 servers) or doubled (600), or London’s capacity is doubled, CARMA’s

Table 5: Sensitivity analysis (spatio-temporal regime; 26 alternating weeks of 2024×2 traces per setting). Oracle red.: emission reduction of the oracle relative to ASAP-Local (%). Gap columns: excess over the oracle (%). Wins: weeks in which CARMA emits less than MPC. Late: share of late work (%), MPC / CARMA. ^aBusy servers draw 0.2 kWh above an idle draw of 0.2 kWh per server-hour that every batch server incurs regardless of the schedule. ^bSchedules are made with average intensities, but emissions are accounted with the stylised marginal emission factor of Section 5.5, and the oracle optimises marginal emissions.

Setting	Oracle		Gap to oracle (%)				Wins vs		Late (%)	
	red.	red.	Greedy	Reserve	MPC	Protect	CARMA	MPC	MPC/CARMA	Late
Urgent share $v = 0.1$	62.0	62.0	79.2	28.5	23.3	16.4	12.0	26/26	0.00 / 0.00	
Urgent share $v = 0.5$	56.4	56.4	53.8	25.5	21.5	15.3	6.7	26/26	0.09 / 0.00	
Network energy $\eta = 0.005$ kWh/GB	62.3	62.3	70.1	28.7	27.3	18.0	10.9	26/26	0.01 / 0.00	
Network energy $\eta = 0.06$ kWh/GB	53.9	53.9	65.5	28.8	19.4	14.9	8.4	26/26	0.00 / 0.00	
Load $\rho = 0.4$, profile learned at 0.5	67.3	67.3	75.9	29.0	23.3	15.9	12.2	26/26	0.00 / 0.00	
Load $\rho = 0.6$, profile learned at 0.5	51.3	51.3	56.0	27.8	24.1	19.0	8.6	26/26	0.31 / 0.00	
Clean-site capacity halved (150)	52.4	52.4	58.2	25.4	21.7	16.9	9.5	26/26	0.02 / 0.00	
Clean-site capacity doubled (600)	69.2	69.2	85.1	32.7	28.6	17.5	9.3	26/26	0.01 / 0.00	
London capacity doubled (1200)	47.6	47.6	53.1	27.9	22.7	18.9	9.5	26/26	0.00 / 0.00	
Always-on servers (idle 0.2 kWh) ^a	18.5	18.5	10.9	4.8	3.6	2.6	1.5	26/26	0.00 / 0.00	
Marginal-emission accounting ^b	13.8	13.8	12.2	10.7	8.6	9.2	7.9	18/26	0.01 / 0.00	

gap stays at 9.3–9.5%. The gaps of MPC (22–29%) and MPC-Protect (16.9–18.9%) stay about two to three times larger. More clean capacity raises the oracle’s reduction (from 52.4% to 69.2%), but a myopic planner still misallocates it.

Misspecified demand profile. CARMA’s ghost jobs come from a profile learned at $\rho = 0.5$. When the actual load is 20% higher ($\rho = 0.6$), so that CARMA under-anticipates, its gap is 8.6% against 19.0% for MPC-Protect and 24.1% for MPC. When the actual load is 20% lower ($\rho = 0.4$), CARMA over-anticipates and reserves more clean capacity than needed. Its gap rises to 12.2%, but it still beats MPC-Protect by 3.7 points (15.9%) and MPC by 11.2 points. The trace experiment of Section 5.4 adds a real profile that underestimates demand by 28%. Because ghost work carries a lower penalty than real work, the planner never plans real work late in order to serve ghost demand. Without a backstop, however, a misprediction can still delay real work once capacity is exhausted (Section 4.2); the late-work column of Table 5 tracks this.

Always-on servers. If every batch server draws idle power whether or not it is busy, a large part of the emissions no longer depends on the schedule. The achievable reduction falls to 18.5% for the oracle and 17.3% for CARMA. The ranking is unchanged (gaps of 1.5% for CARMA, 2.6% for MPC-Protect and 3.6% for MPC). Carbon-aware scheduling is therefore most valuable when it is combined with right-sizing, so that deferred work actually reduces the number of powered servers [18].

Marginal emissions. All sites are in one GB market, where gas-fired plant is usually the marginal generator [7, 46]. Moving load between regions then mainly changes *which* region is assigned the average mix, and it abates emissions physically only when low-carbon output would otherwise be curtailed. We therefore account emissions with a stylised marginal emission factor (MEF). The MEF is 394 gCO₂/kWh (combined-cycle gas) in every hour except two cases where it is zero. For South Scotland it is zero when Scotland is nearly carbon-free (below 25 g) while GB-wide intensity exceeds 100 g, a proxy for export-constrained hours; this holds in 27% of hours. For all sites it is zero when GB-wide intensity is below 50 g (4% of hours). Under this accounting the picture changes substantially. The oracle cuts marginal emissions by only 13.8%, and CARMA by 7.5%. The forecast-driven policies are within 7.9–10.7% of the marginal oracle. CARMA is still the best of them,

881 but its advantage over MPC shrinks to 0.6 points (block-bootstrap CI 0.1–0.9;
882 $p = 0.003$ after Holm correction; CARMA better in 18 of 26 weeks). The
883 schedules were optimised for average intensity, not for the MEF, and the MEF
884 itself is a proxy with fixed thresholds that has not been validated against
885 dispatch or curtailment data. Under this proxy, most of the attributional
886 reductions reported above would be reallocations within the GB system.
887 The physical benefit depends on whether shifting absorbs otherwise-curtailed
888 low-carbon output or changes the marginal generator between hours, and a
889 scheduler should be driven by a validated marginal signal when that is the
890 goal.

891 5.6. Robustness to prediction errors

892 We test how CARMA degrades as its predictions deteriorate on every
893 fourth test week of 2024 (13 weeks, $\rho = 0.5$, migration allowed, fresh workload
894 traces). Controlled errors are injected into the two kinds of prediction (Fig. 7).
895 These experiments do not satisfy the assumptions of the bound in Theorem 2,
896 so they probe the degradation profile empirically.

897 *Carbon-forecast error.* With exact forecasts, CARMA’s gap is 3.8% and
898 MPC’s 22.5%. Independent noise is comparatively benign: CARMA’s gap
899 rises smoothly to 4.5%, 6.3%, 9.7% and 16.1% at $\sigma = 25, 50, 100$ and
900 200 gCO₂/kWh. Noise that persists over a whole target day, as when a
901 weather front is missed, is more damaging. The gaps are then 6.5%, 8.8%,
902 15.8% and 21.7%, because the planner cannot average out the error by re-
903 planning. MPC deteriorates from 22.5% to 30.5–32.0% at the largest noise
904 level. At every noise level and type, CARMA remains better than MPC with
905 the same forecasts. Only with the most persistent large errors ($\sigma = 200$) does
906 it approach the level of MPC with *perfect* forecasts. Fig. 7a places the GBM
907 forecaster at the noise level with the same realised MAE (31.8 gCO₂/kWh,
908 $\sigma \approx 46$). With the GBM forecasts, CARMA’s gap on the same weeks is
909 9.9%, larger than with either noise model at that error size (about 6% and
910 8.5%). The forecaster’s errors, which are both serially and cross-sectionally
911 correlated and concentrated in volatile hours, are therefore more damaging
912 than either stylised noise model, and the degradation should not be read as
913 merely i.i.d.

914 *Demand-prediction error.* Scaling the learned demand profile by g produces a
915 V-shaped response with its minimum at the unbiased profile ($g = 1$, gap 9.2%).

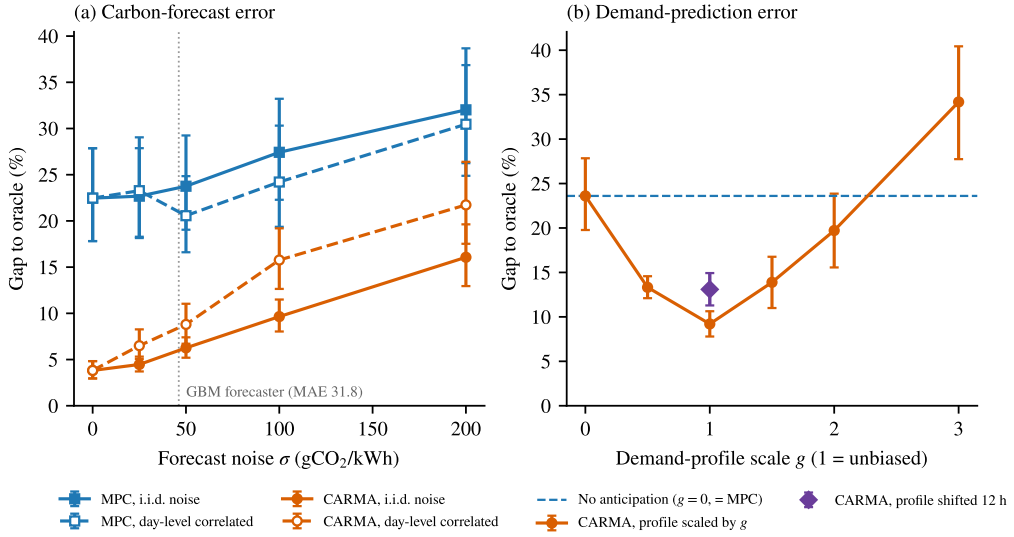


Figure 7: Degradation under controlled prediction errors (13 test weeks of 2024, $\rho = 0.5$, migration allowed; means with 95% confidence intervals). (a) Carbon-intensity forecasts replaced by the realised values plus Gaussian noise of standard deviation σ . The noise is either independent across issue times and horizons (solid) or shared by all forecasts of the same site and target day (dashed, open markers). (b) Demand profile scaled by g , or shifted by 12 hours so that arrivals are predicted at the wrong time of day.

Under-prediction by half ($g = 0.5$) and over-prediction by half ($g = 1.5$) cost 4.1 and 4.7 percentage points. Predicting arrivals at the wrong time of day (a 12-hour shift) costs 3.9 points. Removing anticipation ($g = 0$) returns the MPC level (23.6%). Gross over-prediction ($g = 3$, three times the true demand) is worse than no anticipation (34.2%), because CARMA then holds clean capacity for work that never arrives. A 12-hour shift keeps most of the benefit, which indicates that much of the value comes from anticipating the *volume* of demand rather than its exact timing. Late work was negligible in all 325 runs (at most 0.01% of work). The largest ratio of CARMA’s emissions to the oracle’s over all runs was 1.58.

5.7. Fair carbon attribution

We evaluate Theorem 3 on the 52 test weeks at $\rho = 0.5$, with the four sites as players. For every week and each of the 15 coalitions we compute the hindsight value $c(C)$, and we also run CARMA with only the coalition’s sites and jobs. Each game is one 216-hour arrival episode per week: the evaluated week plus 24 hours of arrivals before and after it. Pooling pays: the hindsight optimum of the grand coalition has a penalised cost of 3.32 tCO₂ per episode, against 6.69 for the four sites operating separately; the emissions alone are 3.32 and 6.58 tCO₂. With CARMA the figures are 3.58 and 6.87. The question is how to share the reduction.

We compare six attributions. The *origin-based* rule charges each site the emissions of its own jobs wherever they run. The *host-based* rule, the analogue of location-based Scope 2 accounting, charges each site the emissions of the work it hosts. The *work-proportional* rule splits the total in proportion to submitted work, and the *stand-alone proportional* rule in proportion to stand-alone emissions. The *Shapley value* [67] averages marginal contributions over all orders of joining. The *dual* rule is Eq. (10). Because $c(C)$ includes the lateness penalty whenever a coalition cannot serve all of its work, we also evaluate an emissions-only game, in which $c(C)$ is the emissions of the hindsight schedule. Stand-alone coalitions leave work unserved in 8 of the 52 episodes (15%), up to 256 server-hours in one episode. In that game the dual attribution is rescaled to the emissions total and is no longer covered by the theorem. For CARMA’s realised emissions, the dual attribution is applied to the hindsight value and the online excess over the hindsight optimum is split in proportion to work. A check is violated when a coalition is charged more than it would emit on its own.

Table 6: Coalitional stability of carbon attributions (52 test weeks of 2024, $\rho = 0.5$; one 216-hour arrival episode per week). Checks: share of the 52×14 (episode, proper coalition) pairs in which the coalition is charged more than its stand-alone value. Weeks: share of episodes with at least one violation. IR: share of (episode, site) pairs in which a site is charged more than its stand-alone value. Neg.: share of (episode, site) pairs with a net credit. Max. exc.: largest violation (tCO₂). Panel (a) uses the penalised values $c(C)$ of Theorem 3, panel (b) emissions only, panel (c) CARMA’s realised emissions.

Attribution	Checks (%)	Weeks (%)	IR (%)	Neg. (%)	Max. exc.
<i>(a) Hindsight, penalised cost</i>					
Origin-based	46.4	100.0	49.0	0.0	1.25
Host-based	49.2	100.0	50.0	0.0	4.30
Work-proportional	44.5	100.0	45.2	0.0	1.43
Stand-alone proportional	22.4	100.0	0.0	0.0	1.05
Shapley	3.6	44.2	0.0	41.3	0.20
Dual	0.0	0.0	0.0	38.0	–
<i>(b) Hindsight, emissions only</i>					
Shapley	3.2	42.3	0.0	41.3	0.13
Rescaled dual	0.0	0.0	0.0	38.0	–
<i>(c) CARMA, realised emissions</i>					
Origin-based	45.3	100.0	45.7	0.0	1.20
Host-based	48.6	100.0	50.0	0.0	4.04
Work-proportional	44.5	100.0	45.2	0.0	1.48
Stand-alone proportional	21.7	100.0	0.0	0.0	1.00
Shapley	3.6	50.0	0.0	41.3	0.12
Adjusted dual (heuristic)	0.5	7.7	0.0	35.6	0.03

952 Table 6 and Fig. 8 report the results. In the hindsight game the dual
 953 attribution satisfies all 728 coalition checks, as Theorem 3 guarantees. The
 954 dual prices are not always unique, but over all optimal duals (checked in 9
 955 episodes) the attribution of a site varies by at most 0.011 tCO₂, so the choice
 956 of dual does not matter in practice. The four accounting rules violate 22–49%
 957 of the checks and fail in every episode. The origin-based rule is individually
 958 irrational for South Scotland in all 52 episodes and for North West England
 959 in 50: their own jobs are displaced to dirtier capacity by imported work, so
 960 they are charged more than they would emit on their own. The host-based
 961 rule fails for the same two sites in every episode, because they are charged for
 962 the imported work they host; North West England, for example, is charged
 963 1.5 tCO₂ per episode against 0.6 on its own. Scaling stand-alone emissions
 964 down is individually rational by construction, but it still leaves 22% of the
 965 coalition checks violated. The Shapley value comes close: it is individually
 966 rational and violates 3.6% of the checks (44% of episodes), with a largest
 967 excess of 0.20 tCO₂. Its site charges are similar to the dual ones. Both credit
 968 the clean sites for the capacity they contribute. Under the dual rule South
 969 Scotland receives a net credit of 0.95 tCO₂ per episode (Shapley 0.80), North
 970 West England is close to neutral, and London’s charge (2.81 t) remains well
 971 below its stand-alone value (3.91 t). Net credits arise in 38% of the (episode,
 972 site) pairs. The Shapley value also needs all 2ⁿ coalition values, whereas the
 973 dual rule comes from a single solve. The emissions-only game gives almost the
 974 same picture: the rescaled dual attribution passes every check, and Shapley
 975 fails 3.2%.

976 With CARMA’s realised emissions, which is the practical case, the adjusted
 977 dual attribution remains individually rational for every site in every episode.
 978 It violates 4 of the 728 coalition checks (0.5%, in 4 of 52 episodes), all
 979 for the coalition of South Scotland, North West England and London, and
 980 the largest excess is 0.03 tCO₂. Shapley applied to the online values fails
 981 3.6% of the checks, and the four accounting rules fail 22–49%. Stand-alone
 982 coalitions under CARMA leave at most 3.7% of their work late, and unfinished
 983 work emits nothing, so the online checks, if anything, favour secession. The
 984 online result is an empirical observation for this heuristic adjustment, which
 985 Theorem 3 does not cover.

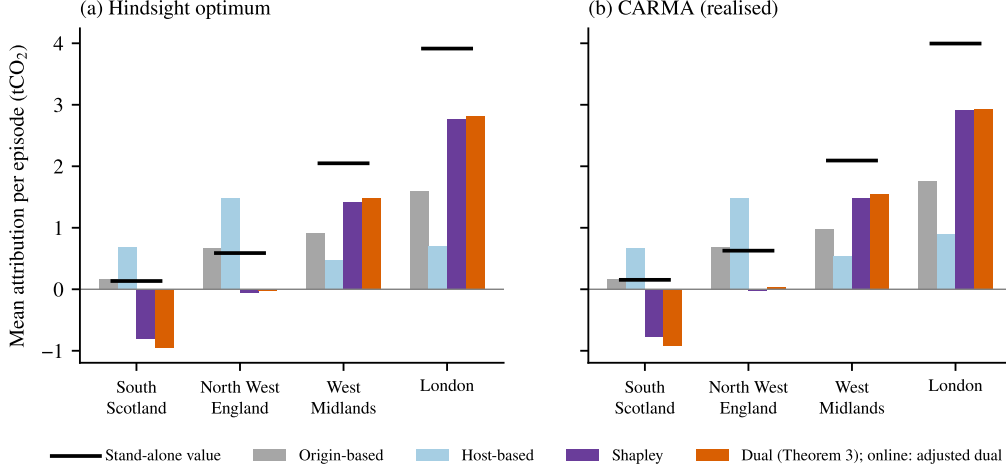


Figure 8: Mean weekly carbon attribution by site under four rules, compared with each site’s stand-alone value (black lines), 52 test weeks of 2024 at $\rho = 0.5$. (a) Hindsight optimum. (b) CARMA’s realised emissions. Negative values are net credits for contributing low-carbon capacity.

6. Discussion

6.1. Managerial implications

Protect scarce clean capacity. In multi-site carbon-aware scheduling, the largest operational loss does not come from imperfect carbon forecasts. It comes from spending low-carbon capacity on work that could have waited or run elsewhere. Proposition 2 and Theorem 1 show this in the worst case, and the experiments confirm it on average. With migration allowed, a myopic planner with perfect carbon foresight is still 17–21% above the optimum in the synthetic experiment and 34% above it on the trace-derived workload. Clean capacity at a low-carbon site should be treated as a scarce resource with an opportunity cost, as in revenue management. A simple protection level (MPC-Protect) captures part of this value. CARMA captures more, because its ghost jobs price the protection hour by hour from the operator’s own history. The result held when the profile underestimated demand by 28% and demand was bursty. A stochastic lookahead over sampled demand scenarios does slightly better still (0.3–1.0 points of gap), at 12–37 times the decision time; when the planning network can be solved in seconds, it is a reasonable alternative.

1004 *Invest in forecasts where they pay.* The value of better intensity forecasts
1005 depends on the rest of the policy. For a myopic multi-site planner, perfect
1006 forecasts are worth only 1.4–2.4 percentage points; once demand is anticipated
1007 they are worth 4–7 points; without migration they are worth almost the whole
1008 remaining gap. Persistent errors, such as a missed day-ahead wind front,
1009 are more costly than independent ones. Forecasting effort, for example the
1010 integration of numerical weather predictions [44, 43], should therefore target
1011 day-level biases, and in multi-site settings it should follow the fix of demand
1012 myopia.

1013 *Use risk premia with care.* The conformal cost views help a myopic planner
1014 but not CARMA. They work by making volatile sites and distant hours look
1015 dirtier, which crudely imitates anticipation. Once demand is anticipated,
1016 the same distortion gives up useful deferrals. Calibrated quantiles are better
1017 suited to reporting and to service-level decisions than to reshaping costs by
1018 default.

1019 *Emissions or service near saturation.* Without migration and at high load,
1020 every carbon-aware policy delayed work, and under a service-adjusted metric
1021 carbon-agnostic operation was preferable. Operators should combine temporal
1022 shifting with explicit service constraints, or restrict it, when local capacity is
1023 close to saturation.

1024 *Attributional versus physical savings.* The headline reductions are attribu-
1025 tional: they measure how much of the work runs where and when the *average*
1026 regional mix is clean. Within a single market such as Great Britain, moving
1027 load between regions mostly reassigns the average mix. Under a stylised
1028 marginal-emission proxy, the achievable reduction fell from 58% to 14% and
1029 CARMA’s advantage over MPC became small (0.6 points). Carbon-aware
1030 shifting is physically most valuable when it absorbs low-carbon output that
1031 would otherwise be curtailed, for example behind transmission constraints
1032 between Scotland and England, when temporal shifts change which plant is
1033 marginal, or when sites lie in different markets. Our marginal accounting is
1034 a stylised proxy, not an estimate of GB-wide abatement, so it indicates the
1035 direction and rough size of this effect rather than its value. In such settings
1036 a marginal signal should drive the scheduler. The scheduling framework is
1037 unchanged: only the cost coefficients of the flow network change. Idle power
1038 works in the same direction. If servers stay powered, deferral saves only

1039 their dynamic energy, and carbon-aware scheduling should be paired with
1040 right-sizing [18].

1041 *Fair internal carbon accounting.* Pooling halved the hindsight emissions of
1042 the four sites. However, origin-based accounting, which charges each site for
1043 its own jobs wherever they run, would have made South Scotland worse off
1044 than on its own in every episode. A pool of organisations or business units
1045 allocated this way is unstable, because its cleanest members have an incentive
1046 to leave. The dual attribution of Theorem 3 charges each site the carbon
1047 shadow price of its demand and credits it for the capacity it contributes. It is
1048 budget-balanced, no coalition would prefer to secede, and it is a by-product
1049 of the scheduling problem. It suits internal carbon pricing and the sharing
1050 of the benefits of carbon-aware scheduling among business units. It is not
1051 a substitute for external GHG Protocol reporting, where negative Scope 2
1052 emissions do not exist [66]. The players must also be entities that could
1053 credibly schedule on their own, such as separate operators, business units
1054 with their own capacity, or tenants with reserved shares. Because credits
1055 reward contributed capacity, capacity declarations should be audited.

1056 *Deployment.* CARMA needs a carbon-intensity feed (average or marginal), a
1057 demand profile computed from the operator’s own job logs and a network-flow
1058 solver. It decides within tens of milliseconds and returns a plan that is optimal
1059 for its model, which makes its decisions auditable. Absolute savings scale
1060 with the fleet: for the 1,700-server fleet simulated here, CARMA saves about
1061 16 tCO₂ per year relative to myopic control and about 165 tCO₂ relative to
1062 carbon-agnostic operation, in attributional terms.

1063 6.2. Limitations and future research

1064 The study has several limitations. First, the synthetic workloads use a
1065 profile learned from the same generator, which favours anticipation. The trace
1066 experiment partly addresses this concern, but it covers only four evaluation
1067 days of one cluster, arrival times are first task starts, work is based on planned
1068 CPU requests, deadlines and origins are synthetic, and jobs are batched. Its
1069 156 episodes vary carbon conditions but share one demand chronology. Second,
1070 the regional intensities published by NESO are modelled estimates, used here
1071 as realised values, 52 hours of them are missing and filled by persistence, and
1072 the marginal analysis relies on a stylised proxy with fixed thresholds, not
1073 on a dispatch model or observed curtailment. Third, the facility parameters

(PUE, network energy, a 96-core server) are stylised, and embodied emissions are outside the scope except as the floor that makes θ finite. Fourth, the forecaster excludes weather inputs, so its day-ahead skill is limited. Fifth, bag-of-tasks jobs with free preemption and migration are an idealisation: jobs with state, data gravity, bandwidth limits or data-sovereignty constraints would add restrictions that break the pure network structure. Sixth, the guarantees of Theorem 2 assume a backstop resource, a positive emission floor and an episode within the planning window, none of which holds in the simulations. The theorem describes the mechanism and the experiments its empirical profile. Its robustness term is shared by all on-time policies and is far from the $\Omega(\sqrt{\theta})$ lower bound. Seventh, the core property is exact only for the hindsight benchmark, and the online adjustment is a heuristic. Eighth, Scenario-MPC was tuned and tested with a modest number of scenarios and was not included in the sensitivity analysis because of its cost.

Several extensions follow naturally. The ghost-demand profile could be replaced by probabilistic arrival forecasts, with the anticipation weight set by a newsvendor-type argument, or by scenario-based planning that treats demand and carbon uncertainty jointly [57]. Tightening robustness towards the lower bound while retaining consistency, for example by protecting deferral decisions with reservation prices [24, 26], is an open theoretical question. The framework can also be driven by marginal emission signals, coupled with right-sizing, on-site renewables and storage [21, 22] and demand-response programmes [23], and evaluated across several electricity markets.

7. Conclusions

This paper studied carbon-aware scheduling of deferrable computing workloads across data-centre sites as a sustainable-operations problem. The offline problem, including width limits, deadlines and the emissions of data transfer, is a minimum-cost flow, which yields exact oracles (after quantisation) and fast online planning. Uncertainty about demand alone forces every online algorithm to a competitive ratio of order $\sqrt{\theta}$, even with perfect carbon forecasts, and myopic planning does far worse. CARMA augments the planning network with ghost jobs for expected arrivals. In an idealised setting with a backstop and episodes inside the planning window, it meets every deadline under any predictions, is optimal under exact predictions and degrades linearly with prediction error; a numerical check in such episodes confirmed

1109 this. The dual prices of the pooled problem yield a carbon attribution that is
1110 budget-balanced and in the core of the hindsight pooling game.

1111 In 52 out-of-sample weeks of 2024 on British regional grid data, with migra-
1112 tion allowed, CARMA cut attributional emissions relative to carbon-agnostic
1113 operation by 37–69%. It came within 8.0–9.8% of the clairvoyant optimum,
1114 against 15.5–16.3% for the best baseline without demand information, and
1115 emitted less than myopic MPC in 51 or 52 of the 52 weeks. A scenario-
1116 based stochastic lookahead with the same demand information was 0.3–1.0
1117 points better but 12–37 times slower. On a trace-derived workload, with
1118 a demand profile that underestimated demand by 28% and bursty arrivals,
1119 CARMA’s gap was 21.2% against 27.5% for the best baseline without demand
1120 information. Anticipating demand was worth far more than perfect carbon
1121 foresight, and correlated day-level forecast errors hurt more than independent
1122 ones. Three results qualify these gains. Without migration, reductions are
1123 small (9–18%), and near saturation carbon-aware shifting delays work enough
1124 that carbon-agnostic operation is preferable under a service-adjusted metric.
1125 Under a stylised marginal-emission proxy, the achievable reduction fell to 14%
1126 and CARMA’s advantage over MPC shrank to 0.6 points, so the physical
1127 value of shifting within one grid depends on absorbing otherwise-curtailed
1128 low-carbon output. With always-on servers, most emissions no longer depend
1129 on the schedule.

1130 Origin-based and host-based carbon accounting violated coalitional stabil-
1131 ity in every test episode. The dual attribution never did in hindsight, and
1132 failed 4 of 728 checks, by at most 0.03 tCO₂, for the heuristic adjustment
1133 to CARMA’s realised emissions. Its charges were close to the Shapley value,
1134 which needs all coalition values, whereas the dual attribution needs one opti-
1135 misation. Carbon-aware computing is most effective when schedulers plan for
1136 the work that has not yet arrived, not only for the grid conditions that lie
1137 ahead, and when its benefits are measured and shared on a sound basis.

1138 CRediT authorship contribution statement

1139 **Quang-Vinh Dang:** Conceptualization, Methodology, Software, Data
1140 curation, Formal analysis, Investigation, Validation, Visualization, Writing –
1141 original draft, Writing – review & editing.

1142 Declaration of competing interest

1143 The author declares no known competing financial interests or personal
1144 relationships that could have appeared to influence the work reported in this
1145 paper.

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1149 Data availability

1150 The carbon-intensity data are publicly available from the NESO Carbon
1151 Intensity API [45] under the CC BY 4.0 licence. The data snapshot used in
1152 this study, all code, the workload generator with fixed random seeds, and all
1153 raw experiment outputs are available from the author’s repository (<https://github.com/vinhqdang/Sustainable-Operations-and-Computers>).
1154

1155 Appendix A. Proofs

1156 *Atoms.* Splitting a job j into sub-jobs $j_1, \dots, j_k$ with work w_{j_i} and widths
1157 $m_{j_i} = m_j w_{j_i} / w_j$ does not change the optimal value of any planning problem.
1158 Given a solution for the merged job, the allocation $x_{j_i s}(t) = x_{j s}(t) w_{j_i} / w_j$
1159 satisfies every constraint of the split problem at the same cost. Conversely,
1160 summing the sub-job allocations gives a feasible allocation for the merged
1161 job, because $\sum_i m_{j_i} = m_j$. Splitting also preserves window-feasibility, since
1162 $w_{j_i} / m_{j_i} = w_j / m_j$. We may therefore compare true and predicted job sets
1163 atom by atom.

1164 Appendix A.1. Proof of Theorem 1

1165 Normalise $L = 1$, so $U = \theta$. Site A has one server with cost $p \in [1, \theta]$ in
1166 hour 0 and cost 1 in hour 1. Site B has ample capacity at cost θ in both
1167 hours. Transfer costs are zero. A flexible job ($w = m = 1$, window $\{0, 1\}$) is
1168 released at $t = 0$. The adversary either releases nothing more (scenario 1) or
1169 releases an urgent job ($w = m = 1$, window $\{1\}$) at $t = 1$ (scenario 2). All
1170 costs are known in advance.

1171 Consider any deterministic online algorithm that may split work fraction-
1172 ally, and let $x \in [0, 1]$ be the amount of flexible work it runs at $t = 0$. Running

1173 at B in hour 0 costs $\theta \geq p$ and is dominated, so we may assume this work runs
 1174 at A . In scenario 1 the algorithm finishes the remaining $1 - x$ at A in hour 1,
 1175 so $\text{ALG}_1 = xp + (1 - x)$, while $\text{OPT}_1 = 1$. In scenario 2, hour 1 must process
 1176 $2 - x$ units and A can take only one of them, so $\text{ALG}_2 = xp + 1 + (1 - x)\theta$,
 1177 while $\text{OPT}_2 = p + 1$. The ratio $r_1(x) = 1 + x(p - 1)$ increases in x , and the
 1178 ratio $r_2(x) = (1 + \theta - x(\theta - p))/(p + 1)$ decreases in x . Hence $\min_x \max\{r_1, r_2\}$
 1179 is attained where they are equal, at $x^* = (\theta - p)/(p^2 - p - 1 + \theta) \in [0, 1]$, and
 1180 its value is

$$1 + \frac{(p - 1)(\theta - p)}{p^2 - p - 1 + \theta}.$$

1181 The adversary chooses p to maximise this value, which gives $R^*(\theta)$. The
 1182 choice $p = \sqrt{\theta}$ gives the stated closed-form lower bound. For a randomised
 1183 algorithm, the expected cost in each scenario is affine in x , so it equals the
 1184 cost of the deterministic fractional algorithm that plays $\mathbb{E}[x]$, and the bound
 1185 applies against an oblivious adversary. Myopic MPC sees only the flexible job
 1186 at $t = 0$ and, for any $p > 1$, defers it entirely to the cheaper hour 1 ($x = 0$).
 1187 Its ratio in scenario 2 is $(1 + \theta)/(p + 1)$, which tends to $(1 + \theta)/2$ as $p \rightarrow 1^+$,
 1188 so its competitive ratio is at least $(1 + \theta)/2$. $\square$

1189 *Appendix A.2. Proof of Proposition 3 and Theorem 2*

1190 *Proposition 3.* A policy that completes all work W on time executes every
 1191 server-hour at a unit cost of at most U , so its emissions are at most UW .
 1192 Every server-hour in the oracle's solution costs at least L , and slack costs
 1193 $\Pi > U \geq L$, so $\text{OPT} \geq LW$. $\square$

1194 *Deadline safety.* We first show by induction that every pending real job j
 1195 is window-feasible at the start of every hour: its remaining work satisfies
 1196 $r_j \leq m_j(d_j - t)$. This holds at release by (A1). Suppose it holds at hour t ,
 1197 and suppose the plan of CARMA leaves some real work $u_j > 0$ unfinished.
 1198 Because $r_j \leq m_j(d_j - t)$, some hour $\tau \in [t, d_j)$ carries less than m_j units
 1199 of job j in the plan. Moving a small amount of u_j to the backstop in hour
 1200 τ keeps the plan feasible, because the backstop has unlimited capacity. It
 1201 changes the planned cost by a negative amount, because the planner prices
 1202 the backstop at most at $U < \Pi$ (A1), whatever its forecasts of the other costs
 1203 are. Since $d_j - t \leq H$ (A1), hour τ lies inside the planning window. This
 1204 contradicts optimality, so the plan completes all real work. After the first hour
 1205 is executed, the rest of the plan is a feasible completion, so $r'_j \leq m_j(d_j - t - 1)$

1206 and the induction step holds. No real job is ever late, whatever the ghost
1207 jobs are, and θ -competitiveness follows from Proposition 3.

1208 *Telescoping.* Let σ_t be the state at the start of hour t : the remaining work of
1209 all real jobs released by t . Let $V_t(\sigma, G; c)$ be the optimal value of (2)–(5) over
1210 hours $t, \dots, T-1$ for the pending jobs σ and the future jobs G under costs c .
1211 Write $\bar{V}_t = V_t(\sigma_t, F_t; c)$, so that $\bar{V}_0 = \text{OPT}$ and $\bar{V}_T = 0$. For a feasible hour- t
1212 allocation x with cost $g_t(x)$ and successor state $\sigma^+(x)$, define

$$Q_t(x) = g_t(x) + V_{t+1}(\sigma^+(x), F_t; c), \quad \hat{Q}_t(x) = g_t(x) + V_{t+1}(\sigma^+(x), \hat{F}_t; \hat{c}_t).$$

1213 By dynamic programming over a deterministic future, $\min_x Q_t(x) = \bar{V}_t$, and
1214 $Q_t(x_t) = g_t(x_t) + \bar{V}_{t+1}$ for the executed action x_t . Summing over t gives

$$\text{ALG} - \text{OPT} = \sum_{t=0}^{T-1} \delta_t, \quad \delta_t = Q_t(x_t) - \min_x Q_t(x) \geq 0.$$

1215 Ghost jobs are released at $t+1$ or later, the planning window reaches T
1216 by (A2), and hour- t costs are observed. The planning problem solved by
1217 CARMA at hour t is therefore exactly $\min_x \hat{Q}_t(x)$, and x_t is a minimiser.

1218 *One-step loss.* Let $x^* \in \arg \min Q_t$. Since $\hat{Q}_t(x_t) \leq \hat{Q}_t(x^*)$, we have $\delta_t \leq$
1219 $\Delta(x_t) - \Delta(x^*)$, where $\Delta = Q_t - \hat{Q}_t$. Write $\Delta = D_1 + D_2$ with

$$\begin{aligned} D_1(x) &= V_{t+1}(\sigma^+(x), F_t; c) - V_{t+1}(\sigma^+(x), \hat{F}_t; c), \\ D_2(x) &= V_{t+1}(\sigma^+(x), \hat{F}_t; c) - V_{t+1}(\sigma^+(x), \hat{F}_t; \hat{c}_t). \end{aligned}$$

1220 *Cost error.* An optimal solution under $\hat{c}_t$ is feasible under c , and its cost
1221 changes by at most ε_t per unit of flow on future-hour arcs. The same holds
1222 with the roles of c and $\hat{c}_t$ exchanged. Hence $|D_2(x)| \leq \varepsilon_t W_t$ for all x , and
1223 $D_2(x_t) - D_2(x^*) \leq 2\varepsilon_t W_t$.

1224 *Demand error.* Under (A1), the penalty attached to a window-feasible
1225 job does not affect any optimal value, provided it is at least U . If a solution
1226 leaves slack on such a job, the argument used for robustness moves that slack
1227 to the backstop at no greater cost. True future jobs (penalty Π) and ghost
1228 jobs (penalty Π_g) can therefore be treated as identical atoms whenever they
1229 coincide. Let a be a window-feasible atom and G any set of future jobs. For
1230 every state σ ,

$$Lw_a \leq V(\sigma, G \cup \{a\}) - V(\sigma, G) \leq Uw_a.$$

1231 For the upper bound, add a to an optimal solution for G by running it on the
 1232 backstop at rate $w_a/(d_a - a_a) \leq m_a$ in every hour of its window. For the lower
 1233 bound, delete a 's flow and slack from an optimal solution for $G \cup \{a\}$; each
 1234 deleted unit cost at least $\min\{L, \Pi_a\} = L$, since $\Pi_a \geq U \geq L$. Now transform
 1235 $\hat{F}_t$ into F_t by removing $\hat{F}_t \setminus F_t$ and adding $F_t \setminus \hat{F}_t$, one atom at a time. Then
 1236 $D_1(x)$ is a sum of increments, each lying in $[Lw_a, Uw_a]$ (with a minus sign
 1237 for removals) for every x . Therefore $D_1(x_t) - D_1(x^*) \leq (U - L) \sum_a w_a =$
 1238 $(U - L) d_t$.

1239 Combining the two bounds gives $\delta_t \leq (U - L)d_t + 2\varepsilon_t W_t$. Summing over t
 1240 and taking the minimum with the robustness bound proves (9). If $d_t = \varepsilon_t = 0$
 1241 for all t , then $\delta_t = 0$ and $\text{ALG} = \text{OPT}$. $\square$

1242 *Appendix A.3. Proof of Theorem 3*

1243 Write the problem of coalition C as $c(C) = \min\{c^\top z : A_- z =$
 1244 $b_-(C), A_\leq z \leq b_\leq(C), z \geq 0\}$, where $z = (x, u)$. Jobs submitted out-
 1245 side C have zero work and zero width, and sites outside C have zero capacity.
 1246 This forces the corresponding flows to zero, so the problem coincides with
 1247 the stand-alone problem of C . The right-hand side is additive over sites:
 1248 $b(C) = \sum_{o \in C} b^o$, where b^o collects the work and widths of the jobs submitted
 1249 at o and the capacity of site o . The matrix A and the cost c do not depend
 1250 on C . The dual problem $\max\{b(C)^\top y : A^\top y \leq c, y_\leq \leq 0\}$ therefore has a
 1251 feasible set that is the same for all coalitions.

1252 Let $y^* = (\lambda, \nu, \mu)$ be an optimal dual solution for $\mathcal{S}$. The slack variables
 1253 make every primal problem feasible, and costs are non-negative, so strong
 1254 duality gives $c(\mathcal{S}) = b(\mathcal{S})^\top y^* = \sum_o (b^o)^\top y^* = \sum_o \phi_o$, which is budget balance.
 1255 For any coalition C , y^* is dual-feasible for C , so weak duality gives $c(C) \geq$
 1256 $b(C)^\top y^* = \sum_{o \in C} \phi_o$, which is the core condition [31]. Expanding $(b^o)^\top y^*$
 1257 gives (10). By Proposition 1 the problem is a min-cost flow, so optimal node
 1258 potentials, together with the reduced costs they induce on saturated capacity
 1259 arcs, also give an optimal dual solution. Any optimal dual solution can be
 1260 used; in our experiments it is returned by the LP solver HiGHS. $\square$

1261 **Appendix B. Forecast evaluation**

1262 Table B.1 and Fig. B.1a summarise forecast accuracy in 2024; hours
 1263 affected by gap filling are excluded. Averaged over horizons and sites, the
 1264 direct GBM forecaster has a mean absolute error (MAE) of 31.7 gCO₂/kWh,
 1265 against 35.3 for persistence and 40.9 for the seasonal-naive forecast. The

advantage is largest at short and medium horizons: MAE falls by 23% at 1 hour (6.4 against 8.3) and by 24% at 6 hours (23.6 against 31.2). The advantage shrinks with the horizon, and at 24 hours the GBM is worse than both benchmarks (42.7 against 40.9). Without weather forecasts, day-ahead changes in wind output are hard to predict from the intensity history alone [44]. By site, the improvement over persistence is largest in absolute terms for the volatile southern sites (West Midlands 50.2 against 57.7, London 38.0 against 43.4), although their MAE is also the highest. In South Scotland, where intensity is low and often flat, persistence is better overall (14.5 against 16.2). Diebold–Mariano tests [68] with absolute-error loss and HAC variance confirm these patterns. For the three English sites, the GBM is significantly more accurate than persistence at horizons of 1–18 hours ($p \leq 0.002$) and indistinguishable at 24 hours ($p \geq 0.09$). For South Scotland it is significantly better at 1 and 3 hours, indistinguishable at 6 hours and significantly worse from 12 hours onwards.

Table B.1: Forecast accuracy on the 2024 test year (MAE in gCO₂/kWh; mean over sites unless stated).

Model	Horizon h (hours)					All horizons	
	1	3	6	12	24	MAE	RMSE
Persistence	8.3	20.0	31.2	38.6	40.9	35.3	49.2
Seasonal naive (24 h)	40.9	40.9	40.8	40.8	40.9	40.9	57.4
Direct GBM (this study)	6.4	15.1	23.6	34.5	42.7	31.7	41.6

MAE by site over all horizons (persistence / GBM): South Scotland 14.5 / 16.2; North West England 25.5 / 22.3; West Midlands 57.7 / 50.2; London 43.4 / 38.0.

Calibration matters more than point accuracy for the risk-adjusted cost view, and Fig. B.1b shows the effect of distribution shift. Static split-conformal bounds, calibrated on July–September 2023, under-cover in 2024 at every level: at $\beta = 0.9$, annual coverage was 0.868 and monthly coverage ranged from 0.78 to 0.95. With the adaptive recalibration, annual coverage is within 0.003 of the target at all levels (0.499, 0.598, 0.697, 0.797 and 0.898). Monthly coverage stays within about four percentage points of the target (at most 4.2 points), and coverage is uniform across horizons. At $\beta = 0.5$, the recalibrated median lies on average 10.3 gCO₂/kWh below the point forecast, which reflects the downward drift in intensity that the static model does not capture. Coverage is not the whole story. In pinball loss, the adaptive bounds improve

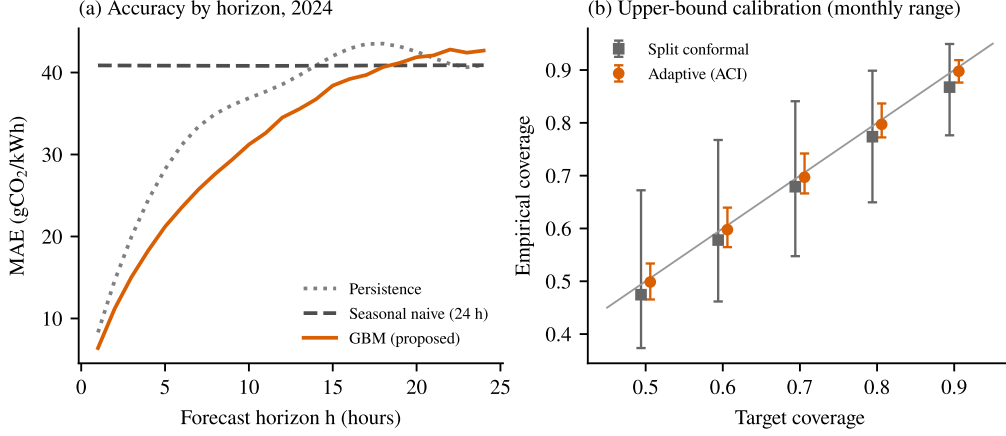


Figure B.1: Forecast evaluation in 2024. (a) MAE by horizon, averaged over sites. (b) Empirical coverage of one-sided upper bounds U^β against the target level β for static split conformal calibration and adaptive conformal inference (ACI). Markers show the annual coverage and bars the range across the 12 months.

on static calibration at levels 0.5–0.7 (15.01 against 15.16 at 0.5) but are worse at 0.8 and 0.9 (9.07 against 8.36 at 0.9). Delayed feedback explains much of this. The level for horizon h is updated only when the outcome is observed, h hours after issue, so a day of under-forecasts drives the level of long horizons towards 1 before any correction arrives. At $\beta = 0.9$, 5.6% of the 2024 bounds (0.5% at $\beta = 0.8$) were at the fallback ceiling, the training-period maximum of the site. These features, together with the finite ceiling and the quantile grid, are why we treat the procedure as an empirical recalibration and do not invoke the coverage guarantee of ACI [54]. Annual coverage says nothing about conditional coverage in the site-hours that the optimiser selects.

Appendix C. Numerical check of Theorem 2

The weekly experiments of Section 5 violate the assumptions of Theorem 2, so we check the theorem separately in episodes that satisfy them (script `verify_theorem2.py`). Each episode is one day ($T = 24 = H$ hours) of the synthetic workload, with days starting at 26 test weeks, a backstop of unlimited capacity at the episode’s largest unit cost U (A1), window-feasible jobs, real-job penalty $\Pi = 10^4 > U$ and ghost penalty $\Pi_g = U$. Ghost jobs are the true future jobs, each omitted from the predictions with probability q

(demand error), and the carbon forecasts of future hours receive i.i.d. Gaussian noise of standard deviation σ gCO₂/kWh (carbon error). For each episode, we compute the clairvoyant optimum OPT with the backstop, the emissions of CARMA, and the right-hand side of the smoothness bound (9) from the realised d_t , ε_t and W_t . There are 26 episodes for each of 12 settings (two loads and six error settings).

Table C.1: Numerical check of Theorem 2 in one-day episodes with a backstop. ALG/OPT: emissions of CARMA relative to the clairvoyant optimum (mean and maximum over 26 episodes). Bound: mean ratio of the right-hand side of (9) to OPT. The bound held in every episode, and no real work was late.

Load ρ	q	σ	Mean ALG/OPT	Max ALG/OPT	Bound / OPT
0.5	0	0	1.000	1.000	1.0
0.5	0	25	1.011	1.031	37.7
0.5	0	100	1.061	1.257	139.6
0.5	0.1	0	1.009	1.072	6.4
0.5	0.3	0	1.066	1.365	17.9
0.5	0.3	100	1.094	1.332	160.2
0.9	0	0	1.000	1.000	1.0
0.9	0	25	1.001	1.003	19.3
0.9	0	100	1.004	1.015	69.3
0.9	0.1	0	1.001	1.011	3.6
0.9	0.3	0	1.010	1.112	9.1
0.9	0.3	100	1.010	1.029	78.2

Three observations follow. With exact predictions, CARMA reproduces the optimum to within the 0.01 server-hour quantisation of the flow solver (relative difference at most 2×10^{-5}), which is consistency. No real work was late in any of the 312 episodes, in line with deadline safety. The smoothness bound holds in every episode but is loose by one to two orders of magnitude when predictions are wrong, because each misprediction enters d_t in every hour in which it is visible. The realised excess over the optimum is at most 37%, so the bound is a qualitative statement that errors enter linearly, not a tight prediction.

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